Ostbayerische Technische Hochschule Amberg-Weiden
Fakultät Elektrotechnik, Medien und Informatik

Studiengang Bachelor Künstliche Intelligenz

**Bachelorarbeit**

von

Sebastian **Weidner**

# Imitation Learning unter Verwendung videokonditionierter Policies für visuell gesteuertes Roboterverhalten

Imitation Learning using Video-Conditioned Policies for Visually Guided Robotic Behavior

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OTH Amberg-Weiden
Elektrotechnik, Medien und Informatik

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Studiengang: **Bachelor Künstliche Intelligenz**

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Ich bestätige, dass ich die Bachelorarbeit mit dem Titel:

**Imitation Learning unter Verwendung videokonditionierter Policies für visuell gesteuertes Roboterverhalten**

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Bachelorarbeit Zusammenfassung

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Titel:

**Imitation Learning unter Verwendung videokonditionierter Policies für visuell gesteuertes Roboterverhalten**

Zusammenfassung:

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Schlüsselwörter: LaTeX, Textsatz, Formeln, Graphiken

Abstract:

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Keywords: LaTeX, Textsatz, Formeln, Graphiken

# Contents

# Symbole, Formelzeichen und Einheiten

# Chapter 1

# Introduction

## 1.1 Motivation

Roboter werden zukünftig eine Schlüsselrolle in zahlreichen Anwendungsfeldern einnehmen. Be-reits heute steigt der Einsatz von Robotersystemen in der industriellen Produktion kontinuierlich an, da diese insbesondere monotone, einfache und repetitive Aufgaben effizient und zuverlässig übernehmen können. Trotz der teilweisen sehr hohen Anschaffungskosten, investieren immer mehr Unternehmen in die Integration von Roboter in Ihre Produktionslinien. Die Vorteile liegen dabei vor allem in der hohen Verfügbarkeit und Planbarkeit: Nach der Inbe-triebnahme fallen nur noch geringe laufende Betriebskosten an, die sich primär auf Energiever-brauch in From von Strom- und Wartungskosten beschränken. Darüber hinaus können Roboter unabhängig von der Arbeitszeit 24h, 7 Tage die Woche betrieben werden und liefern eine gleich-bleibende Qualität. Dies ermöglicht eine hohe Prozessstabilität sowie eine signifikante Steigerung der Produktivität. Im Gegensatz zu menschlichen Arbeitskräften, werden Roboter nicht krank, nehmen nicht an einen Gewerkschaftsstreik teil und es müssen auch keine Sonderzahlungen in Form von Nacht-, Wochenend- und Feiertagszuschlägen geleistet werden. Grundsätzlich lassen sich Robotersysteme hinsichtlich ihrer Programmierung in zwei Kategorien einteilen. In der industriellen Praxis dominieren derzeit statisch programmierte Roboter, die für klar definierte Aufgaben in strukturierten Umgebungen eingesetzt werden. Sie haben feste Bewe-gungsabläufe und können sich nur in geringem Maße mithilfe integrierter intelligenter Sensoren an ihre Umgebung bzw. an die jeweilige Aufgabe anpassen. Solche Systeme sind besonders effi-zient in standardisierten Produktionsprozessen, stoßen jedoch an ihre Grenzen, sobald sich die Umgebung oder die Anforderungen ändern. Im Gegensatz dazu gewinnen intelligente Robotersysteme zunehmend an Bedeutung. Diese sind in der Lage, auf Veränderungen in ihrer Umgebung zu reagieren und ihr Verhalten adaptiv anzu-passen. Insbesondere durch den Einsatz von Künstlicher Intelligenz eröffnen sich neue Möglich-keiten, komplexe und unstrukturierte Aufgaben zu bewältigen. Zentrale Herausforderungen bestehen jedoch hierdabei in der robusten Ausführung vielfältiger Szenarien sowie in der Fähigkeit, auf neue Objekte, Umgebungen und Aufgaben zu generalisieren. Die heute eingesetzten intelligenten Robotersysteme, bzw.

deren zugrunde liegenden KI-Modelle sind oftmals stark auf ihre jeweilige Aufgabe und die Zielumgebung spezialisiert, bzw. trainiert worden. Aufgaben, die nicht im Trainingsdatensatz enthalten sind, können häufig nicht zuverlässig ausge-führt werden. Ebenso treten Probleme auf, wenn gelernte Aufgaben in grundlegend verschiedenen Umgebungen ausgeführt werden sollen. Der Grund hierfür liegt in der Art und Weise, wie die zugrunde liegenden KI-Modelle trainiert werden: Die Trainingsdaten sind in der Regel stark auf eine spezifische Aufgabe und Umgebung zugeschnitten. Eine Erhöhung der Datenvielfalt führt dabei häufig zu einem Zielkonflikt, da sie mit einer geringeren Genauigkeit bei der Ausführung einzelner Aufgaben einhergehen kann. Selbst wenn man ein großes generalisierendes Roboter KI-Modell entwickeln möchte, stellt der Mangel an geeigneten und vielfältigen Roboter-Trainingsdaten ein gravierendes Problem dar. Klassische, aufgabenspezifische Trainingsansätze stoßen hierbei aufgrund mangelnder Skalierbar-keit und Flexibilität an ihre Grenzen. Aktuelle Forschungsansätze verfolgen daher das Ziel, Robo-tern ein visuelles und semantisches Aufgabenverständnis zu vermitteln, sodass sie Handlungen aus hochdimensionalen Eingaben wie Videos oder Bildern ableiten können. Dadurch können neue Datenquellen für das Training des Roboters herangezogen werden. Mit einem visuellen und se-mantischen Verständnis können beispielsweise (menschliche) Videos von YouTube im Training verwendet werden. Diese beinhalten viele verschiedene Handlungen und Ausführungsformen ver-schiedenster Aufgaben, die ein generalisierendes Roboter KI-Modell brauchen würde. Imitation Learning ist ein vielversprechender Ansatz in diesem Kontext, der in dieser Arbeit genauer unter-sucht wird.

Beim Imitation Learning geht es darum, den Roboter seine Aufgabe mithilfe eines (menschlichen) Demonstrationsvideos beizubringen (z.B. das Stapeln von Würfeln, oder das Greifen von Objek-ten). Beim traditionellen Ansatz muss ein Roboter auf die jeweilige Aufgabe mit viel Aufwand bzw. mit vielen Roboter trajectory Datensätzen trainiert werden.

## 1.2 Project Goal

Ziel dieser Arbeit ist es eine Einführung in das Thema Robotic mit Imitation Learning zu liefern. Die Arbeit gliedert sich in zwei Teile. Dabei werden zuerst die Grundlagen der Robotik erläutert und anschließend technisch in das Thema Intelligente Robotik mit dem Fokus Imitation Learning eingeführt. Abgerundet wird der erste Teil mit einem Literaturüberblick, über ausgewählte Imitation Learning Ansätze, die dort näher untersucht werden. Im zweiten Teil dieser Arbeit wird ein Open-Source Imitation Learning Modell experimentell auf einen realen Roboterarm implementiert. Die Erstmalige Inbetriebnahme des Roboters und das Setup, wie ein KI-Modell zur Ausführung auf den Roboter gebracht werden kann, gehört auch dazu. Ziel ist es, den Roboter zu befähigen, einfache Manipulationsaufgaben, beispielsweise das Stapeln von Würfeln, allein auf Basis menschlicher Demonstrationsvideos zu erlernen und auszu-führen. Die Leistungsfähigkeit des vorgeschlagenen Ansatzes wird anschließend experimentell auf dem realen Robotersystem untersucht und evaluiert.

Beim Imitation Learning geht es darum, den Roboter seine Aufgabe mithilfe eines (menschlichen) Demonstrationsvideos beizubringen (z.B. das Stapeln von Würfeln, oder das Greifen von Objek-ten). Beim traditionellen Ansatz muss ein Roboter auf die jeweilige Aufgabe mit viel Aufwand bzw. mit vielen Roboter trajectory Datensätzen trainiert werden. Beim Imitation Learning geht es darum, den Roboter seine Aufgabe mithilfe eines (menschlichen) Demonstrationsvideos beizubringen. Dabei nimmt eine Kamera ein Video von der jeweiligen Person auf, die eine bestimmte Aufgabe ausübt. Anschließend wird dem Roboter-KI-Modell das Demonstrationsvideo der Aufgabe gezeigt, welches wiederum den Roboter entsprechend steuert, um die jeweilige Aufgabe in seiner Umgebung auszuführen.

## 1.3 Research Focus – Imitation Learning

# Chapter 2

# Basics Classical Robotics

## 2.1 Erster Abschnitt von Kapitel 2

test test test test

### 2.1.1 Ein Unterabschnitt

**Ein Unterabschnitt**

jetzt gehts

# Chapter 3

# Basics Intelligent Robotics

## 3.1 Machine Learning Basics

### 3.1.1 Mathematical Notation of formulas

**Capital letters** are used for random variables, e.g. $S$ represents the set of all states in which the robot can be and $A$ represents the set of all possible actions it can perform.

**Lowercase letters** are used for specific values of these random variables, e.g. $s$ stands for a specific robot state and $a$ for a specific action that the robot performs. A lowercase letter is usually followed by a time index $_t$, representing the specific action taken by the robot at time step $t$. In some cases the time index is omitted for simplicity, e.g. $a$ instead of $a_t$ and the next state or action is then denoted as $s'$ or $a'$ instead of $s_{t+1}$ or $a_{t+1}$.

**Capital letters in bold** represents Matrices, e.g. the matrix $\mathbf{X}$.

**Lowercase letters in bold** are used for specific values of (random) variables that are vectors, e.g. $\mathbf{x}$ represents a specific vector.

### 3.1.2 Glossary

**Agent & Policy:**

An agent is an entity that learns to perform a specific behavior through interaction with its environment. In the context of this work, the agent corresponds to the robot, which has to execute a given action or task. The policy can be interpreted as the brain of the agent, determining which action to take next in a given situation in order to achieve the task objective. In our example, the policy controls the robot's motion and actions.

Formally, a policy can be represented as

$$\pi_\theta(a \mid s) \tag{3.1}$$

which defines a probability distribution over actions $a$, conditioned on the current state $s$. In other words, the policy specifies how likely the agent is to select a particular action given its current observation of the environment.

In the classical meaning, the action $a$ corresponds for example to a robot command, such as moving the gripper to a target position $(x, y, z)$ or setting a joint angle to a specific value with a defined velocity. The state $s$ represents the agent's observation, which may include, for example, a sequence of recent camera images (e.g., the last four frames) as well as additional sensor inputs such as proprioceptive signals.

**Robot Trajectory:**

A robot trajectory describes how the robot moves from a starting state to a target state, including position, orientation and, if applicable, speeds, accelerations, and forces acting on the robot. A robot tracetory is a temporal sequence of actions and the resulting robot states.

**Annotation:**

Annotation refers to the process of labeling data. Data is provided with additional information. For example, class affiliations can be determined, target objects can be marked in images, or states and actions can be marked in robot trajectories. Annotated data, or labeled data, form the basis for supervised learning approaches.

Annotating refers to the process of labeling data, in which data is enriched with additional information. For example, this may include assigning class labels, marking target objects in images, or labeling states and actions within robot trajectories. Annotated, or labeled, data forms the foundation for supervised learning approaches.

**Foundation Model vs. Multi-Model Model**

### 3.1.3 Core Machine Learning Categories

There are various approaches to training an AI model for a robot. These involve methods from the four core machine learning categories, which can also be combined with one another:

**Supervised Learning:**

In supervised learning, a model is trained using labeled training data. This means that, for the input data, the output data to be predicted are known. The goal is to train a model that correctly predicts the output data and makes reliable predictions on unseen data.

**Unsupervised Learning:**

In contrast to supervised learning, unsupervised learning uses unlabeled data for model training. The model tries to recognize structures, patterns or relationships in the data on its own.

**Semi Supervised Learning:**

Semi-supervised learning is a combination of supervised and unsupervised learning. In this approach, a model is trained using a small amount of labeled data and a large amount of unlabeled data. The goal is to leverage the advantages of both approaches, particularly when annotating data is time-consuming or costl-intensive. This is relevant in robotics, as the data used in that field is often difficult to label.

**Reinforcement Learning:**

Reinforcement learning (RL) differs fundamentally from the previously mentioned approaches. Instead of learning from fixed datasets, an agent actively interacts with its environment and tries to find out an optimal strategy to reach a defined goal. Through its actions, it influences the state of the environment and subsequently receives feedback in the form of rewards or penalties, see Figure 3.1 for a short overview. In the context of robotics, the robot is the agent. The objective is to learn a strategy (= policy) that maximizes the cumulative reward over time for the given task.

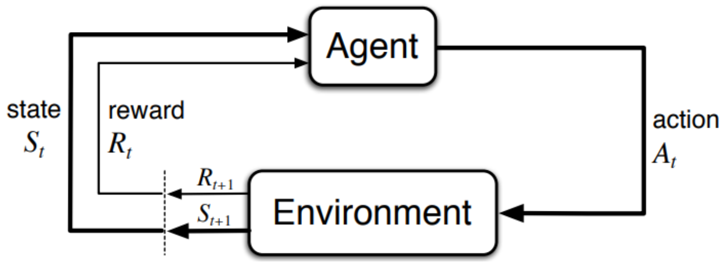



**Figure 3.1:** Base concept of Reinforcement Learning

The agent proceeds in its environment according to the trial-and-error principle to learn an optimal policy. Safety is a critical concern during training, as the agent may execute unpredictable actions while exploring the state space. For this reason, simulation environments are a central component of training and are frequently used for pre-training the policy. The policy is then subsequently often fine-tuned in the real-world environment. Because the agent learns through trial and error, more efficient, unknown, and creative solutions can be found that a human might not have come up with. See section 3.2.1 for more details.

## 3.2 Intelligent robots with Deep Learning (RL)

In general, modern robot training distinguishes between two main approaches: imitation learning and reinforcement learning. Each approach uses a different strategy for training the robot model. In Figure 3.2 an overview of the two approaches is given. The following sections explain the two approaches in more detail and provide a mathematical description of the underlying problem.

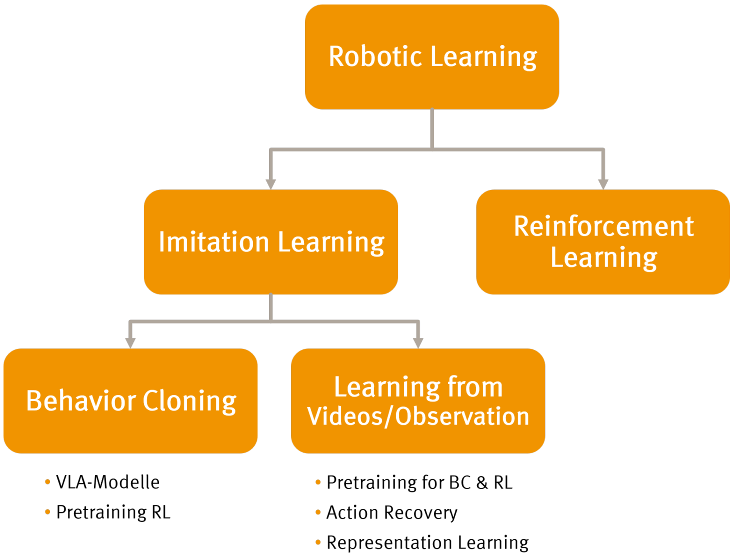



**Figure 3.2:** Overview Robotic Learning

### 3.2.1 Reinforcement Learning

Reinforcement learning (RL) is both a fundamental discipline of machine learning and a distinct approach within Robot Learning. The robot learns a policy through trial and error using rewards provided from the given task and the environment. The actions performed by the agent are evaluated with a reward score, and this score is then used to gradually optimize the policy.

Key components of the RL problem The following section explains the key concepts in RL related to robotics, see also figure 3.1 for a general graphical overview.

- **Agent:** The agent is the entity that learns to perform a specific behavior through interaction with its environment. In the context of this work, the agent corresponds to the robot, which has to execute a given action or task.
- **Environment:** The environment is the setting in which the agent (the robot) operates. Simply put, the environment is the robot's workplace, where it is deployed. In addition to the robot itself, it includes all relevant objects with which the robot interacts. The environment can be either the real world or a simulated environment in which the robot is trained.
- **Robot Actions:** The Robot actions are simply the control commands the robot can send to its actuators, e.g. open or close the gripper, move the arm to position $X, Y, Z$, or the joint angles of the individual robot axes, etc.
- **Reward:** The reward is a numerical value that indicates how well the robot is performing in relation to the task objective. For example, if the robot successfully picks up an object, it might receive a positive reward, while if it fails or drops the object, it might receive a negative reward. The goal of Reinforcement Learning is to learn a policy that maximizes the cumulative reward over time.
- **Robot State:** state is a complete description of the situation in the environment

at a particular point in time. The robot state represents the current situation of the robot and its environment. It can include various types of information, such as the robot's position, orientation, joint angles, sensor readings, and even visual input from cameras [132]. The state is crucial for the robot to make informed decisions about which actions to take.

- **Observation:** An observation is the information that the robot receives from its sensors at a given time step. It may not provide a complete picture of the environment, and thus may not fully represent the true robot state. In our case, the observation is mostly equivalent to the state [132].
- **Policy:** The policy can be interpreted as the brain of the agent. The policy determines which action should be taken next when the agent is in a particular state in order to achieve the task objective. The Policy controls the robot's motion and actions.

**Mathematical description of the RL problem:**

Die RL Modularitäten werden wie folgt dargestellt:

- Action: $a$
- Reward: $r$
- State: $s$
- Observation: $o$
- Policy: $\pi$
- Time step: $t$, e.g. $a_t$ is the action taken at time step $t$, $a_{t+1}$ is the action taken at the next time step, etc.
- $s'$: The next state after taking action $a$ in state $s$, i.e. $s' = s_{t+1}$, $s = s_t$.

<u>Markov Decision Process</u>

Reinforcement Learning is based on the Markov decision process (MDP). A MDP is a stochastic decision-making process that formally models the transition probabilities from a state $s$ to a subsequent state $s'$ provided that action $a$ is performed, expressed mathematically as $P(s'|s, a)$ [118, 75].

The objective is to learn a policy $\pi(a|s)$ that maximizes the expected long-term cumulative reward, defined as $\mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right]$. The discount factor $\gamma \in [0, 1]$ determines the importance of future rewards compared to immediate rewards. A value of $\gamma$ close to 0 makes the agent focus more on immediate rewards, while a value close to 1 encourages the agent to consider long-term rewards more heavily [118, 46].

An MDP also defines a reward function that specifies the reward $r$ an agent receives when taking action $a$ in state $s$ and transitioning to a subsequent state $s'$. This is commonly expressed as $R_a(s, s')$ [118, 75].

The **Markov property** says that all information from the past is contained in the current state, mathematically expressed as $P(S_{t+1}|S_t) = P(S_{t+1}|S_1, S_2, ...S_t)$. In other words, the transition probabilities depend only on the current state and not on the history of previous states [46].

In practice, the Markov property is often not strictly satisfied. For example, a single image captured by a camera does not provide sufficient information to infer the dynamics of the environment. From one image alone, it is not possible to determine whether a car is moving forward at 100 km/h or 10 km/h, reversing, or remaining stationary. However, such information is crucial for making correct decisions in applications like autonomous driving [117]. Therefore, in practice, only sufficiently informative state representations are often used because the ideal Markov property is not strictly fulfilled. This is often achieved by incorporating a history of observations, such as a sequence of the last five camera frames, to better capture the underlying dynamics of the environment [75].

Training process:

Zu Beginn des Trainingsprozesses ist die Policy in der Regel zufällig, was bedeutet, dass der Roboter zunächst zufällige Bewegungen ausführt. Durch Ausprobieren versucht der Roboter nun Aktionen zu finden, die mit einer hohen Belohnung einhergehen. Diese Aktionen werden anschließend von der Policy höher gewichtet, was bedeutet, dass der Roboter in Zukunft eher dazu neigt, diese Aktionen auszuführen. Für eine erfolgreiche Lernphase ist es wichtig, dass der Roboter eine gute Balance zwischen Exploration (Ausprobieren neuer Aktionen im jeweiligen Zustand) und Exploitation (Nutzung bereits bekannter, erfolgreicher Aktionen im jeweiligen Zustand) findet. Auf die Exploration wird zu Beginn des Trainingsprozesses mehr Wert gelegt, um die Umgebung besser kennenzulernen und die effizientesten Aktionen im jeweiligen Roboter State für den Erfolg der Aufgabe zu finden. Im Laufe der Zeit, wenn der Roboter mehr über die Umgebung und die Belohnungsstruktur gelernt hat, wird die Exploitation wichtiger, um die bereits gefundenen erfolgreichen Aktionen häufiger auszuführen und so die Gesamtleistung, die am Aufgabenerfolg gemessen wird zu verbessern. Es existieren zahlreiche Methoden und Algorithmen, wie die Exploration und Exploitation Phasen effektiv gesteuert werden können. Beispielalgorithmen sind $\epsilon$-greedy, Upper Confidence Bound (UCB), Thompson Sampling. Im Rahmen der Arbeit wird aber nicht näher darauf eingegangen.

Durch die Explorationsphase wird erst spät der Erfolg des gewählten RL-Ansatzes sichtbar, anders als beim Klassischen Deep Learning, siehe Abbildung 3.3.

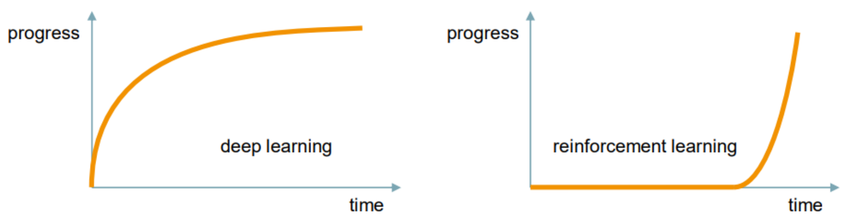



**Figure 3.3:** Training Progress in Deep Learning and Reinforcement Learning

**Knowledge Modality**

An RL Knowledge Modality (KM) is some function learned from data that represents specific types of RL-related knowledge. KMs can be learned through trial-and-error principles or from various data sources, like video datasets, and can capture different aspects of the RL problem.

**Knowledge Modalities (KMs) from Video**

- `Policy:` A Policy $\pi(a_t|s_t)$ is a mapping from states to actions and defines the behavior of the agent that is in a particular state. $\pi(a_t|s_t)$ is the probability of taking action $a_t$ given that the agent is in state $s_t$. $t$ denotes the time step. In other words, the policy specifies how likely the agent is to select a particular action given its current observation of the environment. It can be learned from video data by observing the actions taken in different states and learning to predict those actions based on the visual input.
- `Value functions:` Value functions estimate the expected future return (cumulative reward) for a given state or state-action pair. They can be learned from video data by observing the outcomes of actions and learning to predict the expected future rewards based on the visual input.
  - `State-value function (V-function):` A V-function $V_\pi(s_t)$ is a mapping from states to the expected future return when acting under a particular policy $\pi$.

    **Intuition:** "How good is it to be a certain state, if I follow my policy $\pi$?"

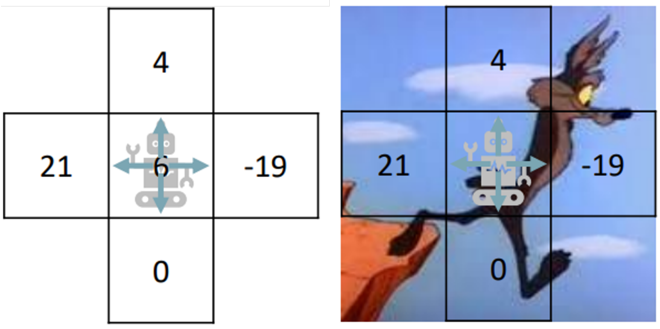



**Figure 3.4:** State-Value Function

  - `State-action value function (Q-function):`

    A Q-function $Q_\pi(s_t, a_t)$ is a mapping from state-action pairs to the expected future return when acting under a particular policy $\pi$.

    **Intuition:** "How good is this specific action here, if I continue behaving like my policy $\pi$?"
  - `V-function vs. Q-function:` Bei der V-function weiß ich nur, wie gut es ist, in einem bestimmten Zustand zu sein, wenn ich nach meiner Policy $\pi$ agiere, aber nicht, wie gut eine konkrete Aktion in diesem Zustand ist. Bei

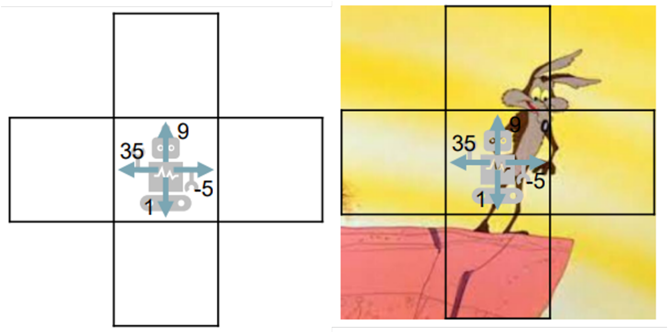



**Figure 3.5:** State-Action Value Function

der Q-function hingegen weiß ich bereits, wie gut die konkrete Aktion $a_t$ im Zustand $s_t$ ist, wenn ich weiterhin nach meiner Policy $\pi$ agiere. Siehe dazu die Abbildung 3.4 und 3.5.

- `Dynamics model:` A Dynamic model $p(s_{t+1}|s_t, a_t)$ predicts the next state $s_{t+1}$ given the current state $s_t$ and action $a_t$. Video prediction models can be used as robot dynamics models. By learning to predict future video frames based on current observations and actions, the model can capture the underlying dynamics of the environment. This allows the robot to anticipate the consequences of its actions and make informed decisions based on predicted future states.
- `Reward model:` A reward function $r(s_t, a_t)$ defines the reward received by the agent for taking action $a_t$ in state $s_t$. It can be learned from video data by observing the outcomes of actions and inferring the underlying reward structure that motivates those actions. For example, if a video shows a robot successfully picking up an object, the reward function can learn to assign a positive reward to that action in similar states.

### 3.2.2 Imitation Learning

Imitation Learning (IL) or Learning from Demonstration is a general term for methods in which the agent derives or learns its behavior from expert demonstrations.

Imitation learning makes it possible to use "in the wild" videos—such as those found in great variety on the internet, for example on platforms like YouTube — to train AI models.

The demonstration videos can show a robot, a human, or even a loose sequence of actions, both in the real world or in a simulation environment, performing a specific task. However, videos without a clearly defined objective or with unknown activity can also be used.

This allows models to be trained to capture the visual and semantic relationships, as well as the physical dynamics of the real world. This, in turn, is a key prerequisite for the development of a generalist policy. Imitation learning can be divided into Behavior Cloning and Learning from Videos/Observation.

**Behavior Cloning (BC)**

Behavior Cloning (BC) is a fundamental approach within imitation learning, where a policy is learned directly from **annotated expert demonstrations**. The training data consists of state–action pairs $(s_t, a_t)$, and the problem is formulated as a standard supervised learning task.

Conceptually, Behavior Cloning can be understood as a form of direct imitation ("direct copying") of expert behavior, without explicitly modeling the underlying system dynamics or task objectives. Due to its simplicity and scalability, BC is one of the most widely used methods in robot learning from datasets.

BC is used to train Vision Language Action models and for pre-training models used in Reinforcement Learning, see Section **XXX**

**Definition:**

Behavior Cloning (BC) learns a policy by supervised learning on expert demonstrations:

$$\pi_\theta(a_t \mid s_t) \tag{3.2}$$

where:

- $s_t$: State / Observation (images, proprioception, etc.) at time $t$
- $a_t$: Expert action at time $t$

The Policy $\pi_\theta(a \mid s)$ is optimized by minimizing the prediction error between the actions generated by the model and the corresponding expert actions.

$$\min_\theta \mathbb{E}_{(s,a)\sim\mathcal{D}} \left[\mathcal{L}(\pi_\theta(s), a)\right] \tag{3.3}$$

where:

- $\pi_\theta(s)$: The action your robot's policy predicts given state $s$.
- $a$: The actual action the expert took in that same state.
- $\mathcal{L}$: Usually Mean Squared Error (MSE) for continuous control or Cross-Entropy for discrete actions.
- $\mathbb{E}_{(s,a)\sim\mathcal{D}}[\cdot]$: Take all the state-action pairs $(s, a)$ inside the expert dataset $\mathcal{D}$, and calculate the average value of the expression inside the brackets.

**Learning from Videos / Observation**

Learning from Videos (LfV), also referred to as Learning from Observation (LfO), is an Imitation Learning approach that relies solely on **video demonstrations without explicit action annotations**. In contrast to Behavior Cloning, where state–action pairs are directly available, LfV aims to infer the underlying actions or high-level intent purely from perceptual data, i.e., mapping observations to actions $(s \to a)$. It often uses *Representation Learning*, to learn and discover the most important (unknown) underlying data characteristics (features) by transforming and reducing input in a fully data-driven and automatic manner.

A central challenge in LfV is the **inference problem**: determining which actions correspond to the observed behavior in the video. Since no action labels are provided, additional methods and techniques are required to recover or approximate the missing action information. See chapter **XXX** for more information.

Due to this property, LfV is typically categorized as a **weakly supervised or self-supervised imitation learning** approach. Its main advantage lies in scalability, as it enables leveraging large amounts of unannotated video data, including internet videos or human demonstrations, without the need for costly data annotation. Furthermore, additional actions can be derived from annotated video data too.

In robotics, Learning from Videos is commonly used as a **pretraining step**. The learned representations capture visual, semantic, and dynamical aspects of the environment, which can later be refined using methods such as Behavior Cloning with annotated data or in Reinforcement Learning. This combination allows models to first learn the real worlds dynamics from large-scale video data and subsequently specialize on task-specific robot demonstrations.

**Definition:**

Learning from Videos (LfV) learns to infer actions from observations in a weakly supervised or self-supervised manner.

$$\hat{a}_t = f(s_t, s_{t+1}) \tag{3.4}$$

where:

- $\hat{a}_t$: The "latent" or predicted action.
- $s_t$: The current state/video frame.
- $s_{t+1}$: The next state/video frame.
- $f$: The inverse dynamics function (often a neural network) that maps state transitions to actions.

### 3.2.3 Summary IL & RL

Reinforcement learning is a powerful approach for learning complex tasks, as it enables agents to improve their behavior through interaction with the environment. However, it can be sample inefficient and may require a large number of interactions to learn an effective policy, especially in high-dimensional state and action spaces. Therefore, it is often combined with other learning paradigms, such as supervised learning or imitation learning, in order to leverage their complementary strengths and improve overall performance.

Imitation Learning generally refers to a family of methods and techniques that enable learning from demonstrations. Two important approaches in this context are Behavior

Cloning (BC) and Learning from Videos (LfV), which differ primarily in the type of data they utilize. BC relies on annotated demonstrations, where state-action pairs are available, and learns a policy directly via supervised learning. In contrast, LfV aims to infer actions from unannotated video observations, where only visual information is provided.

Methods from the LfV paradigm can be used for Action Recovery, making unannotated video data applicable for Behavior Cloning by recovering or estimating the underlying actions. Furthermore, such methods can be employed to pretrain models for BC or Reinforcement Learning, for example by learning expressive state and action representations through Representation Learning.

Behavior Cloning is also widely used for training Vision-Language-Action (VLA) models and for pretraining Reinforcement Learning Knowledge Modalities, for instance by learning components like value functions or reward models. In practice, BC and LfV are often combined to exploit the advantages of both approaches, enabling the use of large-scale unannotated data while maintaining the efficiency of supervised policy learning.

# Chapter 4

# Imitation Learning fundamentals



Was sollte noch erklärt werden?

Online Learning / Offline Learning

Multi-Modal und Foundation Models

## 4.1 Why Imitation Learning?

### 4.1.1 Idea behind Imitation Learning:

When we want to learn a new skill, we often turn to How-To videos. We learn the new skill through speech instructions and visual observations in the video.

`Advantages of Imitation Learning:`

Visual IL ...

- is ideal for teaching tasks that are difficult or impossible to describe in words.

- allows robots to infer tasks from expert demonstrations in different environments than the robot.
- can be more data efficient than reinforcement learning, especially in high-dimensional state spaces, like a complex robot environment.
- can leverage large datasets of human demonstrations, which are often easier to collect than expert demonstrations in the specific domain.

`Problems in RL:` A Task in RL is commonly defined through a reward function, which assigns a high reward to desirable behavior and a low reward to undesirable behavior. However, designing good reward functions is challenging for every task, and many methods based on computer vision rely on annotated demonstrations in every new environment.

IL allows us to learn complex behaviors that are difficult to specify through a reward function, or it can help us to define a good adaptive reward function for RL.

Robotic specific problems: Gap between videos of humans and robots performing similar tasks Gap between simulation and Real-World Simulations & Trial and Error in real World possible Limitations: time and computational consuming, Safety?

### 4.1.2 Ways to instruct robots

This section discusses the various ways in which the robot can be taught its task and how the interaction between humans and robots can be designed.

Scene Understanding / Intent Inference

The robot derives its task solely from the environment and the objects present in it [48].

`One Policy, Multiple Tasks:` In this setting, a single policy is typically specialized on a finite set of task instances, as they must be explicitly trained for those tasks. Tasks that are not represented or underrepresented in the training dataset, can therefore often not be executed reliably [15, 71]. For example, if the policy has been trained on multiple manipulation tasks and one of them involves pouring liquid from a teapot into a cup, the policy will execute this behavior when a matching scene is observed during inference [48]. However, generalization to other objects or situations is often limited. Using different teapots or cups may not be problematic as long as the relevant visual features are recognized. In contrast, performance typically decreases under substantial domain shift (significantly different objects or scenes) [48]. If a corresponding pouring behavior from a pot was not included during training, the policy may fail to transfer the action to that new object category [15, 48]. A major limitation of this approach is task ambiguity caused by overlapping object context across different behaviors. If the same object types, such as a teapot and a cup, appear in both a pick-and-place task and a pouring task, the policy may not be able to disambiguate the correct action target based on scene context alone [119].

`One Policy, One Task:` An alternative approach is to train a separate policy for each individual task, which is then selected explicitly during inference. The task-specific policy selection can be performed either manually by a human or automatically by a separate AI classification model that analyzes the scene and determines the appropriate task-specific policy [23]. For example, one policy could be trained for “pouring”, another for “opening a drawer”, and another for “setting up an overturned object”. This modular design is closely related to task-aware expert routing and decoupled planning-and-execution frameworks in robotic manipulation [23].

This decoupling enables training each policy on significantly larger and more diverse datasets containing varied object appearances and environmental conditions, while keeping the core task semantics fixed [97]. Consequently, these specialized policies often support more robust generalization across novel object instances and environmental variations compared to the `One Policy, Multiple Tasks` paradigm. [6, 79, 119]

Language instructions:

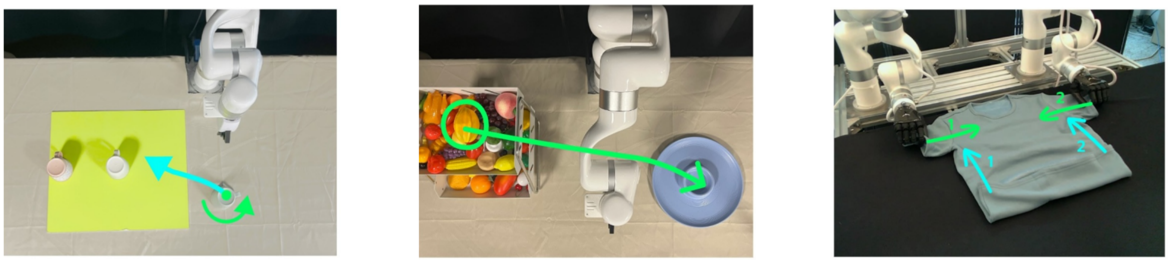
For humans is the most comfortable and natural way to instruct the robot with voice commands. This is very useful for simple tasks that are easy to describe. But complex tasks, where execution must follow exact instructions, are sometimes difficult to describe in words. For example, how would you teach a humanoid robot to decorate tables exactly like a sample table, especially when various decorations need to be assembled in a complex arrangement? Furthermore, ambiguities may arise, and the same instruction may be interpreted differently in different linguistic contexts. For example, the word “open” is used for: open drawer, open cabinet, open container with lid or open jar with screw cap. Although the same verb is used here, each interaction requires a completely different motor control.

A big advantage to command the robot with voice commands is that it enables an interactive dialogue between humans and the robot. If the robot does not understand the instructions, it can ask follow-up questions to clarify any misunderstandings. This helps minimize the problems associated with voice commands.

Image instructions:

Robots can also be instructed using visual inputs such as images. Compared to language, image-based instructions provide a more concrete and unambiguous specification of spatial arrangements, object relations, and desired outcomes.

A common approach is the use of *goal images*, where the robot is given a visual representation of the desired final state. In this setting, the task is formulated as a goal-conditioned problem: the robot observes the current state and acts such that it minimizes the difference to the goal image. This is often referred to as goal-conditioned behavior. However, this approach does not explicitly specify *how* the task should be executed, only *what* the final configuration should look like. But some works have addressed this issue with the usage of annotations in the image of the initial state to specify the task execution. For example [6] highlights the relevant objects and regions with circles and arrows to visualize the task execution in the image.



(a) Task: Rotate cup and place it on table (b) Task: Put fruit into bowl (c) Task: Fold T-shirt

**Figure 4.1:** Image-based instructions for task execution [6].

Another way to address this limitation, *subgoal images* or intermediate states can be provided [86]. These guide the robot through a sequence of visually defined steps, effectively decomposing complex tasks into simpler stages. This is particularly useful for long-horizon or multi-step manipulation tasks.

A key advantage of image instructions is their ability to capture fine-grained spatial details that are difficult to describe in language, such as precise object placement, orientation or symmetry alignments. However, they also introduce challenges related to perception, such as viewpoint variation, occlusions of objects in the real environment, or the use of in-the-wild goal-images with varying lighting and background conditions.

<u>Video-Based Instructions:</u>

Demonstration videos are a powerful way to instruct robots, as they contain rich visual and temporal information. In contrast to single images, videos provide not only the desired goal state of a task but also the sequence of intermediate steps required to reach that goal.

In this setting, the task is typically formulated as an imitation learning problem, where the robot learns a policy that maps visual observations to actions by imitating the demonstrated trajectory. The temporal structure of the video enables the robot to infer *how* a task should be executed, rather than only *what* the final outcome should look like. Depending on the setup, this may also require inferring latent actions from visual observations alone.

Ideally, the robot model should be able to understand it's task from in-the-wild demonstration videos of both robots and humans. In-the-wild videos refer to video data collected in unconstrained, real-world environments, without controlled settings, standardized viewpoints, or curated conditions. Such data exhibits large variability in lighting, backgrounds, and camera motion, making it significantly more challenging for perception and learning compared to controlled laboratory data.

`Human Video Demonstrations:`

- **Data abundance:** In this paradigm, the robot learns from videos of humans performing a task. This is particularly appealing, as large amounts of human demonstration data are readily available (e.g. online videos).

- **Embodiment gap:** A core challenge is the mismatch between human and robot bodies. Human motions and kinematics differ significantly from those of robots. Therefore, the robot must learn to map observed human actions to its own action space.
- **Suboptimal demonstrations:** Another key challenge of human video demonstrations is that they do not necessarily represent optimal task executions. Demonstrations often reflect individual preferences, inefficiencies, or context-specific behaviors, resulting in suboptimal trajectories. In many cases, a video captures only a single possible way of solving the task, without providing information about alternative or more efficient strategies. However, the objective for the robot is typically to perform tasks efficiently and robustly. Therefore, the learned policy must be able to generalize beyond the demonstrated behavior and, if necessary, improve upon it, rather than directly imitating potentially suboptimal actions.

`Robot Video Demonstrations:`

- **No embodiment gap:** The demonstrations are performed by the robot itself or by another robot with similar embodiment. This avoids the human-robot embodiment gap and provides more directly transferable data.
- **Cost and effort:** However, collecting such demonstrations is often expensive and time-consuming, as it requires access to robotic systems and typically controlled environments. In addition, purely visual robot demonstrations may still lack explicit action labels. This means that the robot must still infer the underlying actions and intentions from the visual data.
- **Value of robot demos:** Despite these challenges, robot video demonstrations can provide valuable data for learning, as they are more closely aligned with the robot's own capabilities and constraints compared to human demonstrations.

`Teleoperation:`

- **Direct data collection:** A common method to generate high-quality robot demonstrations is teleoperation, where a human operator directly controls the robot, e.g., via a joystick or a dedicated control interface.
- **Precision and safety:** In contrast to passive video demonstrations, teleoperation provides access to both the visual observations and the corresponding action and trajectory data. As a result, the demonstrated actions are already expressed in the robot's action space, which significantly simplifies reproducing the expected behavior. Such demonstrations are particularly important for fine-grained manipulation tasks that require precise control and accurate timing.
- **Scalability issues:** However, teleoperation may require specialized hardware (controllers, sometimes motion-capture or VR setups) and does not scale as easily as collecting passive video data. Thus, collecting large teleoperation datasets can be time-consuming and is typically done in lab settings only.

## 4.2 Generalist Robots

The ultimate goal of robotics is to develop a `generalist robot`. According to McCarthy et al. [72], such a robot should be able to perform a wide variety of physical tasks in unstructured, unseen real-world environments. It must combine high-level reasoning and planning abilities with low-level physical skills, such as dexterous manipulation. Furthermore, it should be capable of adapting to unforeseen situations and operate under imperfect and partial observations, for example through visual and tactile sensing. In summary, a general-purpose robot is capable of adapting to different situations and learning new skills just like humans [128].

Such a robot would be highly useful across a wide range of real-world applications. In `industrial settings`, this would enable robots to be deployed flexibly across different environments and tasks, without the need for task-specific or environment-specific programming or retraining. As a result, robots could also be utilized for spontaneous small, short-term tasks, where automation is currently not economically viable due to high setup costs and effort. This increased flexibility would lead to higher utilization rates and, consequently, reduced long-term operational costs [128].

In addition, general-purpose robots could play an important role in `household` settings. They could assist with everyday tasks such as tidy up rooms, (un)loading dishwashers, setting tables, or acting as general assistants for example helping to carry objects.

In `healthcare and assistive contexts`, such robots could support individuals with limited mobility by helping them perform daily activities, such as standing up, sitting down, opening a door, or transporting objects. This highlights the potential of generalizing robotic systems to improve both efficiency in industry and quality of life in everyday and care settings [128].

**Ethical concerns:** Ethical concerns regarding the development of generalist robots should not be underestimated and must not be ignored. In the future, robots are likely to take over many tasks, particularly those that are repetitive and monotonous, which may lead to the displacement of certain jobs. Whether this will result in large-scale unemployment due to automation remains a subject of ongoing debate.

Furthermore, once the technology has sufficiently matured, such robots could also be used for `military purposes`. While this could reduce the risks to human soldiers, it also raises serious ethical concerns, as it makes forms of automated or autonomous warfare increasingly conceivable.

It is therefore essential that the development and deployment of generalist robots are carried out responsibly and in accordance with ethical guidelines, in order to minimize potential negative impacts on society.

### 4.2.1 Data Bottleneck in Robotics

A central question in robotics is how to train a generalist robot capable of performing a wide range of tasks in unstructured and previously unseen real-world environments?

`Usage of large-scale datasets.` Progress in related machine learning domains—such as Natural Language Processing [1], Computer Vision [10], and video generation [18]—suggests that large-scale datasets and model capacity are key drivers of performance and generalization capabilities. Training high-capacity models with expressive architectures on diverse, extensive internet-scale datasets has led to the emergence of robust representations and improved out-of-distribution performance [130]. Empirically, these improvements often follow predictable scaling laws, where performance improves as more data, model size, and compute are used [50]. Recent work suggests that similar scaling behavior may also apply to robotics and imitation learning. For example, studies show that robots generalize better wehn they are trained on more diverse training data, including variations in environments, objects, and tasks [63, 24]. Moreover, large-scale imitation learning approaches based on behavioral cloning have demonstrated that training a single policy across many tasks can yield substantial gains in robustness and generalization. For example, transformer-based robot policies such as RT-1 and RT-2 leverage large, heterogeneous datasets to achieve zero-shot generalization to novel tasks and instructions [16, 17].

`Data bottleneck.` In contrast to vision and language domains, robotics is fundamentally constrained by the cost and complexity of data collection, limiting the availability of large, diverse, and high-quality datasets. High-quality demonstrations require physical real world interaction, making them time-consuming to acquire, and are often limited in diversity due to hardware and safety constraints [128]. As a result, available datasets are usually smaller and less diverse. Consequently, this data scarcity represents a major bottleneck for achieving similar levels of scalability and generalization in robot learning, and remains a central challenge in the development of general-purpose robotic systems [128].

### 4.2.2 Relevant Data Categories for Robotics

### 4.2.3 How to Address the Data Bottleneck

How can we address this data bottleneck in robotics? Before discussing specific approaches, it is important to clarify which data sources are relevant for robot learning and how such data can be acquired.

- `Real world data` Real-world robot data is obtained through physical interaction between the robot and its environment and represents a high-quality source for robot learning. Through direct interaction, trajectories and sensor measurements can be recorded under real operating conditions, capturing effects such as sensor noise and environment variability that are difficult to model accurately in simulation.

  `Teleoperation.` A common approach to collecting such data is human teleoperation, where an operator controls the robot to perform tasks and generate demonstration trajectories [36, 129, 195]. This type of data has been widely used in imitation learning and has contributed to the development of early robot foundation models [36, 264]. Teleoperation can be realized through various control interfaces, including joysticks, motion-capture systems, or virtual reality setups. Teleoperation enables precise control and is particularly suitable for collecting demonstrations of complex manipulation tasks that require fine-grained movements and accurate timing. However, it is time-consuming and often requires specialized hardware, which limits scalability. To mitigate this, datasets are increasingly aggregated across multiple research institutions, improving both scale and diversity [195].

`Data collection improvements.` To further address the limitations of manual data collection, recent work has explored automated or semi-automated data acquisition methods, where robots collect data with reduced human intervention [35, 5, 304]. In parallel, advances in robotic infrastructure, particularly in industrial settings, indicate that large-scale data collection in real-world environments is becoming increasingly feasible [251, 115].

Despite these developments, real-world datasets remain significantly smaller and less diverse compared to internet-scale datasets used in vision and language domains. This limitation continues to pose a challenge for scaling robot learning methods and achieving robust generalization.

- `Simulated data`

  `Simulatedly Generated Demonstration Data.` Simulation enables the generation of robot trajectories in virtual environments without the cost, safety risks, and hardware limitations associated with real-world experiments. As a result, simulated demonstrations can be used in scenarios where collecting physical robot data would be impractical, expensive, or unsafe [336, 127, 190]. These generated trajectories are commonly used in imitation learning especially in Behavior Cloning, where policies are trained on demonstration.

  Expert demonstrations in simulation are typically generated by executing a high-performing controller or policy in the simulated environment, such as a hand-crafted controller, a planning-based system, or a policy obtained via reinforcement learning.

  – **hand-crafted controllers:** Hand-crafted controllers are based on classical control methods such as PD/PID controllers, Linear Quadratic Regulators (LQR), or Model Predictive Control (MPC) [93, 76]. These methods are particularly effective for low-dimensional tasks, including balancing, setpoint tracking, or waypoint following, where they can generate stable and accurate trajectories. However, they often struggle in complex or high-dimensional environments.

  – **rule-based and planning-based systems:** For manipulation and navigation tasks, motion planning algorithms such as A*, PRM, or RRT* can be used to compute collision-free trajectories. Frameworks such as MoveIt [77] integrate these planning methods with low-level robot controllers and are widely used in robotics applications [55]. These approaches can generate high-quality demonstrations but often require accurate environment models and can become computationally expensive in complex scenarios.

– **policies obtained via reinforcement learning:** Another approach is to train a reinforcement learning policy, froze it and use it to generate expert demonstration trajectories for downstream imitation learning. Compared to hand-crafted or planning-based approaches, reinforcement learning policies can better adapt to high-dimensional and complex tasks, although training them typically requires large amounts of interaction data and computational resources.

  In simulation, these policies can additionally be trained using *privileged simulation information*, which would not be accessible in real-world settings. Examples include exact object positions, velocities, segmentation masks, or the full environment state. Access to this additional information can simplify the learning process and improve the quality of the generated trajectories.

  Another opportunity is to simplify the problem by training separate expert policies for each subtask. For example, train one policy for grasping, one for navigation, and one for the pulling motion for opening drawers and coordinate them to form a complete task.

`Online Learning in Simulation.` In addition to generating demonstration datasets, simulation is widely used for online learning in robotics, particularly in reinforcement learning. In online learning, the robot continuously interacts with the environment and improves its policy based on newly collected experience during training. Simulation is especially suitable for this process because it enables safe exploration and automatic resetting of environments, parallelized data collection, and faster-than-real-time execution without risking damage to physical hardware. Consequently, simulation has become an essential component for scalable robot learning systems that require large amounts of interaction data, particularly in RL [336, 127, 190].

`Sim-to-Real Gap.` Despite its advantages, simulation-based robot learning faces several important challenges [25]. One major issue is the sim-to-real gap, which arises from inaccuracies between simulated and real-world physics. Effects such as friction, contact dynamics, deformable objects, latency, and sensor noise are often difficult to model accurately. This gap can lead to poor performance when transferring policies trained in simulation to real robots [329]. A common strategy to mitigate this problem is domain randomization, where environmental properties such as textures, lighting conditions, physical parameters, and sensor characteristics are randomized during training to improve robustness and transferability [266].

`Task Diversity.` Another limitation is the difficulty of creating sufficiently diverse simulated tasks and environments for general-purpose robot learning. Manually designing large numbers of environments is labor-intensive and does not scale efficiently. Recent work therefore explores procedural environment generation, where environments and task configurations are generated automatically to increase diversity and improve policy generalization [62, 262, 292, 76, 286].

Despite these challenges, simulated demonstrations remain a valuable source of training data for imitation learning, as they enable scalable trajectory generation and rapid experimentation in controlled environments.

- `Internet data.` Internet data is a promising source for robot learning and includes various types of freely available data, such as text, images, videos, and structured information. Its main advantage lies in its scale and diversity, which can improve the robustness, adaptability, and generalization capabilities of learned robot policies. In analogy to recent advances in natural language processing and computer vision, robotics can similarly benefit from leveraging vast, weakly labeled data from the web.

  In particular, these data sources can support robot learning in several complementary ways. For example, image and video datasets have been used to pretrain vision models that provide robust visual representations for robotic perception systems. In addition, Vision-Language Models (VLMs) and Large Language Models (LLMs) have been employed to define reward functions and task objectives for robot learning. LLMs have also been used as high-level planners for long-horizon tasks by decomposing complex goals into smaller executable actions. Furthermore, several recent works adapt internet-pretrained foundation models for robotics through fine-tuning on robot-specific datasets, while others jointly train on both internet data and action-labeled robot trajectories to improve generalization and cross-domain transfer capabilities.

  In this work, the focus lies on internet video data, which is discussed in more detail in the following sections.

`Combination of data sources.` In modern robot learning systems, simulation data, real-world data, and internet data are rarely used in isolation. Instead, these data sources are commonly combined, as each category provides complementary strengths and addresses different limitations. Simulation enables scalable and cost-efficient data generation and provides a safe environment for online robot interaction and reinforcement learning [89, 159, 89, 80]. Internet data offers large-scale and diverse semantic information [24, 17], while real-world data captures accurate physical interactions, sensor characteristics, and real-world dynamics that are difficult to model in simulation [80].

Typical robot learning pipelines therefore combine these data sources hierarchically [17, 83]. In many approaches, perception and policy models are first pretrained on large-scale simulation and/or internet datasets in order to learn robust and generalizable representations. Subsequently, the models are fine-tuned on smaller but high-quality real-world robot datasets to adapt them to specific robotic platforms, sensors, and target environments [24]. Some works additionally employ hybrid training strategies that combine simulated and real-world data using techniques such as domain randomization or domain adaptation to further reduce the gap between simulated and real-world observations [122, 47].

### 4.2.4 Reduce the amount of robot data

To train a generalist robot, large amounts of diverse data are required in order to capture the variability of real-world environments and tasks. In particular, robot interaction data, also known as robot data, is highly valuable because it contains both observations of the environment, such from RGB-D cameras or other sensors, and the corresponding robot trajectories and actions. This enables the robot to learn the relationship between observations, actions, and their resulting outcomes.

However, the currently available amount of robot data is still insufficient in both scale and diversity to train highly generalizable robotic systems. Collecting robot data, whether in real-world environments or in simulation, is expensive and time-consuming. Real-world data collection requires physical hardware, human supervision, and extensive interaction time, while simulation environments often suffer from the sim-to-real gap and still require significant engineering effort. As a result, data scarcity remains one of the central challenges in the development of general-purpose robotic systems.

**Large-scale internet video data.** To adress this data bottleneck, one promising approach is to leverage large-scale internet video data. Internet videos provide a rich source of information about human behaviors, object interactions, and task execution in diverse environments that are relevant for robotics. They contain fundamental information about physical dynamics, a wide variety of scenarios and interactions (motion patterns), and semantic task structure that can be useful for robotic learning. In contrast to robot datasets, internet-scale video data is abundant, inexpensive to obtain, and covers a much broader range of scenarios and activities as robot datasets.

Although such videos typically do not contain direct robot actions, recent work has shown that they can still provide strong supervisory signals for representation learning, action understanding, and policy pretraining. By augmenting traditional robot datasets with large-scale video data, robotic systems can learn more robust and transferable representations, improving their ability to generalize to unseen tasks and environments [63].

`Formalsim.`

- $D$: The complete training dataset used for learning the robotic policy. This dataset is a combination of the categories real-world data, simulated data, and internet data. $D$ can be subdivided into two subsets: $D = \{D_{robot}, D_{video}\}$.
- $D_{robot}$: Subset of dataset $D$ containing robot interaction data collected from physical robots in real-world or simulation environments, including observations and the robots corresponding actions obtained during interaction. Each sample in $D_{robot}$ consists of transition tuples of the form $(o_t, a_t, r_t, o_{t+1})$, where $o_t$ is the observation at time $t$, $a_t$ is the action taken by the robot, $r_t$ is the reward received after taking action $a_t$, and $o_{t+1}$ is the resulting observation after executing action $a_t$. The reward $r_t$ may be missing, and each $o$ contains an image observation and can additionally contain other information, such as depth or tactile data.

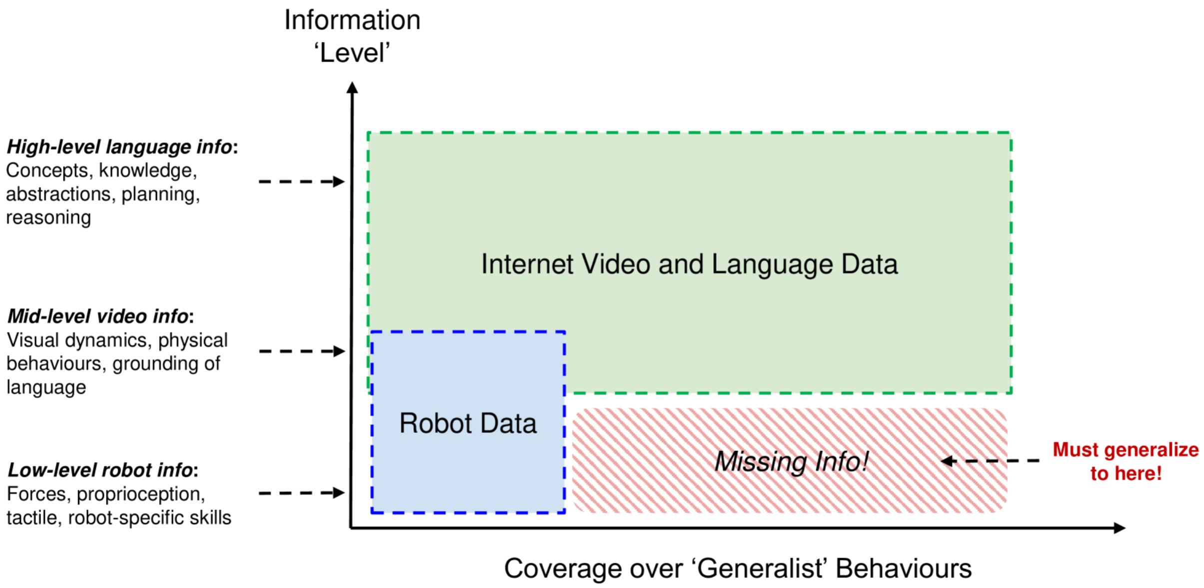



**Figure 4.2:** Problem: Data Bottleneck in Robotics. Source: [73].

- $D_{video}$: Subset of the dataset $D$ that consists the video (demonstration) data. $D_{video}$ can be obtained from external sources such as internet videos. A video clip is represented as $\tau = (o_0, o_1, ..., o_T)$, where $\tau$ denotes the full video clip and each observation $o_t$ corresponds to an RGB image. Depending on the dataset, videos may additionally be paired with language descriptions, action annotations, or other auxiliary modalities.

Figure 4.2 illustrates the data bottleneck in robotics. The x-axis represents the spectrum of behaviors that a generalist robot is expected to perform, while the y-axis reflects the different levels of information present in various data sources. The figure highlights that, although internet data encompasses a broader range of behaviors in diverse real-world scenarios compared to specialized robot datasets, it often lacks the detailed, low-level information that is critical for robotic applications. Ideally, for every relevant behavior, both low-level robot data and internet data would be available, allowing a model to establish connections across different information levels. However, as illustrated in Figure 4.2, there is a significant lack of low-level robot data for many behaviors that are well represented in internet data (see "Missing Info" area). This absence of fine-grained information poses a significant challenge for generalizing robot learning beyond the limitations of available robot datasets. Overcoming this gap and effectively leveraging both broad internet data and precise robot data is a central objective in advancing LfV approaches [73].

`Goal of Learning from Video (LfV).` LfV aims to exploit this abundance of video data to improve robotic learning. [73] says, the goal of LfV is to leverage our combined LfV dataset, $D = \{D_{robot}, D_{video}\}$, to obtain an improved policy $\pi(a_t|s_t)$ versus when learning from $D_{robot}$ alone.

`Improvements of LfV.` What can we expect from combining both data sources, $D_{\text{robot}}$ and $D_{\text{video}}$, instead of using only $D_{\text{robot}}$?

`Generalization beyond` $D_{\texttt{robot}}$ One important improvement of Learning from Video (LfV) is the potential to generalize beyond the limited set of tasks contained in the robot dataset $D_{\text{robot}}$. While robot data primarily provides information about low-level robotic skills, such as grasping or locomotion, internet-scale video data contains a much broader variety of tasks and human behaviors. By combining both data sources, models may learn how low-level robot actions relate to higher-level task structures observed in videos. As a result, an LfV system may be capable of solving tasks that are not explicitly present in $D_{\text{robot}}$ but are represented within $D_{\text{video}}$.

`Emergent Capabilities.` In natural language processing and computer vision, the availability of large-scale internet data has led to significant improvements and, in some cases, the emergence of capabilities that were not explicitly trained for. Similar progress is expected in robotics. By combining robot interaction data with internet-scale video data with language annotations, models may learn to connect low-level robotic skills with rich semantic abstractions and higher-level world knowledge obtained from textual data. This combination has the potential to improve generalization, task understanding, and the ability to solve previously unseen tasks.

`Improvements in-distribution of` $D_{\texttt{robot}}$`.` Furthermore, video data may also improve performance on tasks that are already represented within the robot dataset $D_{\text{robot}}$. Since robot demonstrations are expensive to collect, additional video data can improve data efficiency and help models learn more robust visual and task representations. As a result, Learning from Video (LfV) approaches may achieve higher task success rates and overall better performance compared to approaches trained solely on robot interaction data.

**Information Levels:** Robot data primarily contains low-level and mid-level information, and can be collected from both real-world and simulation, with real-world data being more valuable. In contrast, internet data is typically richer in mid-level and high-level information. The following sections explain each information level in detail.

- `Low-level robot information.` Low-level robot information refers to sensory and control signals that directly describe the robot's physical state, motion, and interaction with its environment. This includes, for instance, end-effector pose, joint positions, joint velocities, joint torques, and gripper state (e.g., open or closed). Additionally, low-level information includes information about the physical forces acting on the robot itself. In simulation environments, they may additionally include physical details of objects in the environment, such as contact forces, object properties or object dynamics. Such low-level information is essential for learning precise motor control tasks, including grasping, locomotion, and manipulation.

  **Proprioceptive information** refers to the robot's internal sense of its own body and movement. It includes data about the robot's end-effector pose which implicity inlcudes joint angles, joint velocities, and gripper configuration. This information is crucial for executing accurate and stable motions, as it allows the robot to understand its current state and adjust its actions accordingly. A robot may visually observe a cup through a camera and wants to grasp it.

Proprioceptive information tells the controller how to move the arm to the cup, which includes:

- what is the current pose/state of the robot
- how far the arm has moved,
- whether the wrist is rotated correctly,
- whether the gripper is aligned with the handle.

**Tactile data** refers to information obtained through physical contact between the robot and its environment. It is comparable to the human sense of touch. Common tactile signals include contact forces, pressure distributions, slip detection, surface texture information, and vibration signals. These measurements are typically acquired using sensors placed on robot grippers, fingertips, or other contact surfaces. For instance, during grasping, tactile sensors can indicate whether an object has been successfully grasped, whether it is slipping, or whether excessive force is being applied. This feedback is critical for maintaining stable and safe object manipulation.

- `Mid-level video information.` Mid-level video information refers to visual observations that capture the dynamics of objects, agents, and environments over time without providing explicit low-level control signals. In contrast to robot data, video data does not include direct information about actions or internal robot states. Instead, it provides sequences of RGB observations that encode how the visual scene evolves over time.

  This type of information includes visual dynamics such as object motion, changes in spatial configurations, and interactions between entities in the scene. For example, videos can capture how objects are moved, grasped, or manipulated by humans, as well as how these actions affect the surrounding environment. Although no explicit force or control signals are available, the physical behavior of objects (e.g., sliding, rotating, or deforming) can often be inferred implicitly from the visual changes across frames.

  A key aspect of mid-level video information is that it encodes **physical behavior at a perceptual level**. While it does not provide direct measurements of forces or torques, it still reflects underlying physical principles such as gravity, friction, and object permanence through observable motion patterns. For instance, a video of a person pouring water into a glass implicitly contains information about liquid flow dynamics and object interaction constraints.

  In addition, mid-level video information provides a natural source for **language grounding**. Many video datasets are paired with textual annotations, captions, or narration, which link visual observations to semantic concepts. This allows models to associate language with observed actions and events, such as "opening a door", "placing an object on a table", or "picking up a cup". Through such alignment, video data helps bridge the gap between perception and high-level task understanding, enabling learning of more abstract representations that go beyond purely visual dynamics.

- `High-level language information.` High-level language information refers to abstract semantic descriptions of tasks, concepts, and behaviors expressed in natural language. In contrast to visual or robot-centric signals, language does not encode direct sensory observations or motor commands, but instead provides structured knowledge about goals, constraints, and task structures. This includes descriptions of concepts (e.g., objects and their properties), relational knowledge, and symbolic abstractions of actions and environments.

  Such information is particularly important for representing task intent and enabling generalization across different embodiments and environments. Language can describe tasks at multiple levels of abstraction, ranging from low-level action descriptions (e.g., "pick up the cup") to high-level procedural or goal-oriented instructions (e.g., "prepare a table for breakfast"). This hierarchical structure allows models to connect concrete behaviors with more abstract task specifications.

  In addition, language plays a central role in **planning and reasoning**. It provides a medium for expressing sequential structure, subgoals, and dependencies between actions. For example, a task such as "clean the table" can be decomposed into intermediate steps such as removing objects, wiping the surface, and reorganizing items. Such decompositions reflect implicit reasoning about task structure and enable more systematic policy learning.

  Furthermore, language serves as a bridge for incorporating prior knowledge into robotic systems. Through textual descriptions, models can access commonsense knowledge about object affordances, physical interactions, and typical human behaviors. This enables improved generalization to unseen tasks by leveraging semantic relationships that are not explicitly present in robot or video data alone.

### 4.2.5 Challenges of Video Data

Although internet-scale video data offers significant potential for robotic learning, it also introduces several fundamental challenges, see Figure 4.3. In contrast to robot interaction datasets, videos are typically high-dimensional, noisy, weakly labeled, highly diverse and lacks on low-level information. Extracting meaningful representations and task-relevant information from such data is therefore a non-trivial problem.

`Missing Action Labels.` One major limitation of raw video data is the absence of explicit action labels required by many imitation learning and offline reinforcement learning approaches. While videos provide rich visual observations of behaviors, they typically do not contain the low-level control commands executed during the task. Retrospectively inferring robot actions from human videos is challenging, since human motions often cannot be directly mapped to the action space of a robotic system. Furthermore, videos usually lack additional reinforcement learning signals such as reward annotations, goal labels, or start-of- and end-of-episode boundaries.

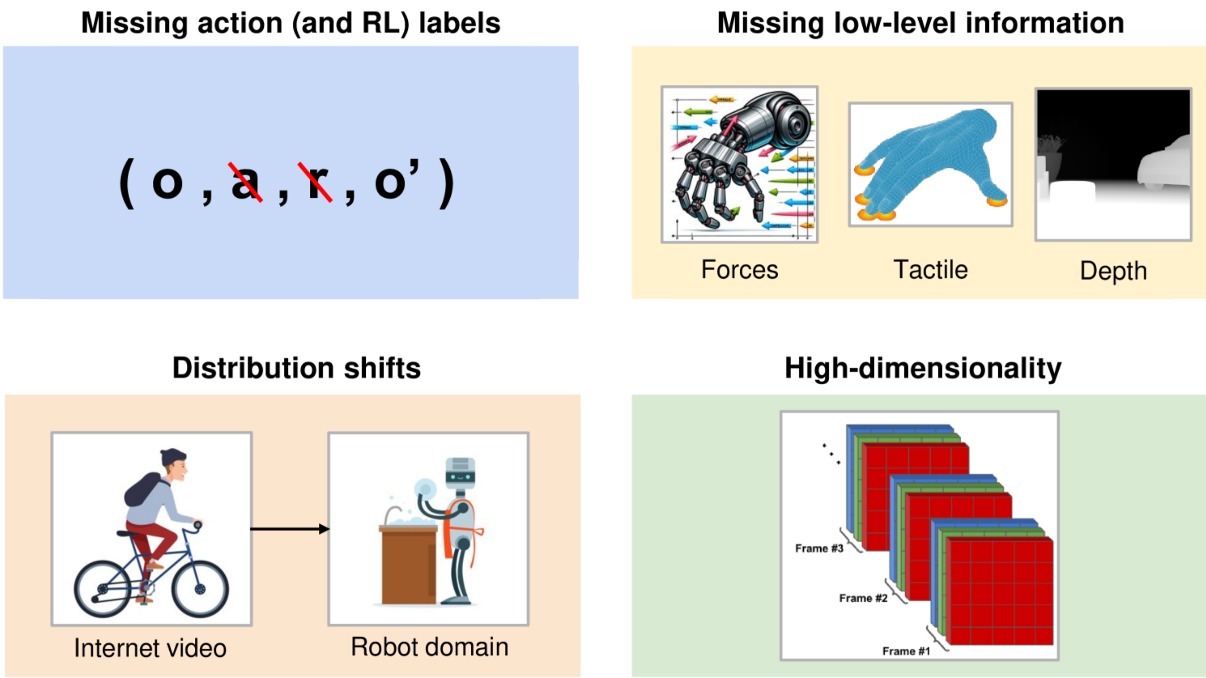



**Figure 4.3:** Key challenges in Learning from Videos [73].

`Missing Low-level Information.` Internet videos also lack on low-level sensory information. Skillful behaviors often depend on tactile feedback, force measurements, proprioceptive signals, or depth information, which are generally not observable in classical RGB video. Although videos may capture high-level task structure and object interactions, they do not explicitly provide the fine-grained physical information necessary for learning accurate motor control and stable interaction with the environment.

`Distribution Shift.` Another challenge arises from the distribution shift between internet videos and the downstream robotics domain. Differences may occur in camera viewpoints, environments, embodiments, object appearances, and task distributions. The Human demonstrations in videos may contain behaviors that are inefficient, ambiguous, or irrelevant for robotic execution.

In particular, Learning from Video (LfV) must address the embodiment gap between humans and robots. Human operators and robotic systems often differ significantly in their kinematics, morphology, and control capabilities. Furthermore, there exists a visual gap between the human actor performing the task in the video and the robotic embodiment that is expected to execute the task. As a result, transferring observed human motions directly to robot actions is highly non-trivial and requires models to learn embodiment-invariant task representations.

Additionally, some important robotic tasks, such as specialized industrial manipulation tasks, may be difficult to learn because they are underrepresented in publicly available video datasets.

`High-dimensionality, Noise, and Redundancy.` Video data is inherently high-dimensional and often contains substantial amounts of irrelevant or redundant information. Background changes, camera motion, occlusions, and unrelated actions can make it difficult for models to identify task-relevant features, making it challenging to learn robust representations. Furthermore, the sheer volume of video data can lead to computational challenges in processing and learning from such large datasets.

`Controllability, Stochasticity, and Partial Observability.` In unlabeled video data, it can be difficult for models to determine which changes in the environment are caused by the agent's actions and which originate from external factors or environmental noise. This ambiguity makes it challenging to infer action information directly from video observations. Furthermore, real-world environments are frequently stochastic and only partially observable, meaning that important state information may be hidden from the camera viewpoint. As a result, predicting future states or accurately modeling action consequences from video alone becomes a difficult problem.

None of these challenges have yet been fully solved. In the following sections, we will discuss some possible ways to address these issues coming with internet video data. Nevertheless it seems likely that scaling — with the help of internet video data — can provide significant gains in robotic systems within the near-future [73].

### 4.2.6 Other Paths to Generalist Robots.

Goal: reduce the amount of robot videos

- **Video provides foundational information:** - Contains physics and dynamics information - Shows human behaviors, tasks and actions relevant to robots - High quantity and diversity - Relatively untapped resource - —>can be highly informative for general-purpose robots. use of diverse internet data has facilitated a transition from task-specific models to monolithic foundation models with more general capabilities

LfV will solve the data bottleneck by augmenting traditional robot data with large-scale internet video.

# Chapter 5

# Learning from Videos - Data Modality and Challenges

Videos are a challenging data modality, as you have seen in Section 4.2.5. This section addresses the two key questions for LfV research:

1. How to extract relevant knowledge from internet video?
2. How to apply video-extracted knowledge to robotic systems for policy learning and control?

We first review different approaches for extracting foundational knowledge from internet video, including video encoders, video prediction models, and video-to-text models. We then discuss methods for transferring video-extracted knowledge to robotics, addressing challenges such as missing action labels and distribution shift. Finally, we explore the intersection of LfV and reinforcement learning, highlighting how video-based learning can enhance RL algorithms and enable more efficient policy learning from visual data.

## 5.1 Extracting Foundational Knowledge from Internet Video

Recent progress in **Video Foundation Models (Video FMs)** provides a promising direction for addressing the first answer: How to extract robotic relevant knowledge from internet video. Video Foundation Models are large-scale deep learning models trained on massive internet video datasets. Through large-scale training, these models learn general visual and temporal representations that capture semantics, motion dynamics, physical interactions, and behavioral patterns. As a result, they can extract foundational knowledge that is transferable across a wide range of downstream tasks, including robotics applications [70].

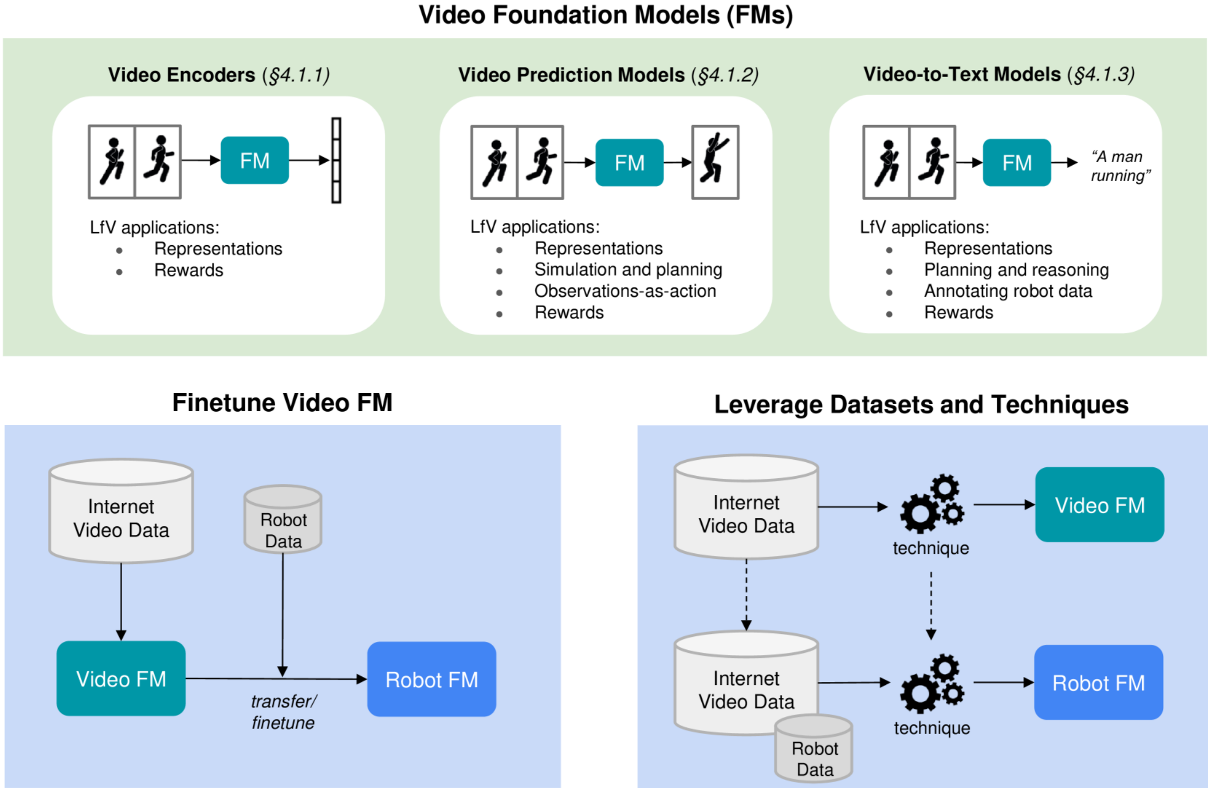



**Figure 5.1:** Video foundation modelling for Learning from Videos [73].

[73] says, Video FMs can be highly applicable for the following LfV use-cases, see Figure 5.1:

- `Using pretrained FMs:` One important application of Video FMs is the transfer of pretrained models to robotics tasks. For example, a video prediction model trained on internet video data may be adapted or fine-tuned into a robot dynamics model, allowing the robot to leverage the broad gathered knowledge about object motion and environmental dynamics from the pretrained Video FM [148, 17, 2, 54].

- `Using techniques and datasets inspired by Video FMs:` In addition, techniques developed for video foundation models can be integrated into robot learning pipelines and can lead to better extraction of representations and knowledge. This, in turn, can lead to improved generalization and task understanding. For instance, a Robot FM can be directly trained by combining robot data with large-scale video inspired by video FMs techniques [120, 16].

### 5.1.1 Video Encoders

Foundational video encoders $p(z \mid \tau)$ are capable of extracting rich and robust representations from raw video observations. Given a video clip $\tau$, the encoder maps the visual input into a latent representation $z$, which captures semantic and temporal information relevant for downstream robotic tasks.

In robotic imitation learning, pretrained video encoders are commonly used in two ways [73]:

- `Representation Learning:` Pretrained video encoders can be transferred to robotic learning frameworks and further fine-tuned on robot-specific data for downstream tasks such as policy learning or an RL Knowledge Modality. The learned latent representations provide compact and informative features that improve generalization and sample efficiency, by allowing the model to learn task-specific features while still leveraging the general knowledge captured from internet video
- `Reward Learning:` Frozen video representations can also be used to define reward functions for robotic agents [81, 115, 35]. By comparing the latent representations of demonstrations and robot executions, the system can measure behavioral similarity without requiring manually engineered reward signals. This approach enables the robot to learn from video demonstrations by optimizing its behavior to match the latent representations of successful demonstrations, thus facilitating imitation learning and skill acquisition.

**Methods**

**General Overview of of Video Representation Learning.**

Video representation learning aims to learn meaningful latent representations from raw video data. These representations capture semantic, spatial and temporal information that can be used for various downstream tasks [73].

`How to learn a video encoder?`

Let

- $X$ denote a video clip
- $\bar{X}$ denote a video clip paired with some corresponding labels (bounding boxes and object masks [28]) and data modalities (language annotations [42], meta-data [103])
- $Y$ denote a learning objective, derived from $\bar{X}$

**Learning goal during training:** predict $Y$ from $X$. Or in other words, learn a video encoder $p(z|X)$ that maps the video clip $X$ to a latent representation $z$ that can be used to predict $Y$: $X \xrightarrow{p(z|X)} z \to Y$ [73].

Basic techniques for learning $Y$ include:

- **Prediction-based Objectives**: The target $Y$, which is derived from the annotated video clip $\bar{X}$, may correspond to future video frames, masked regions, motion information, or other supervision signals. For example, in video prediction tasks, the model predicts the next frame $X_{t+1}$ from the raw previous observations $X_{\leq t}$ [37, 49].

- **Joint-Embedding Objectives**: Joint-embedding approaches [129, 139, 8] first map both the video input $X$ and the target $Y$ into a shared latent embedding space:

$$z_X = f(X), \qquad z_Y = g(Y).$$

A loss function is then applied to align semantically similar embeddings.

$$\mathcal{L}_{\text{pred}} = L_{\text{align}}(z_X, z_Y)$$

Contrastive learning is one of the most widely used approaches in this setting [103].

**State-of-the-Art Approaches**

Recent work has proposed scalable methods for learning rich spatio-temporal video representations.

`Video-text objectives` leverage language annotations to learn semantic representations from video data. Common approaches include:

- **Video-Text Contrastive Learning:** This approaches learn to align video representations with corresponding text descriptions. By maximizing the similarity between video and text embeddings, the model captures semantic relationships between visual content and language [144, 158, 137, 84, 60].
- **Video-Text Matching:** This methods train the model to determine whether a given video clip and text description correspond to each other. The model learns to distinguish between matching and non-matching pairs, which encourages it to capture relevant features for both modalities [60].
- **Masked Language Modelling:** In this approaches, the model learns to predict masked words in a text description based on the corresponding video content. This encourages the model to learn contextual relationships between visual information and language [60].
- **Video-to-Text Generation:** This method trains the model to generate descriptive text based on video input. By learning to produce coherent and relevant descriptions, the model captures high-level semantic information from the video data [84, 146].

`Auto-encoding approaches` have also become highly effective for self-supervised video representation learning. Auto-Encoding Approaches learn video representations by reconstructing input data from compressed latent representations. Common approaches include:

- **Masked Auto-Encoding (MAE):** In this approach, parts of the video input are randomly masked, and the model is trained to reconstruct the missing regions from the visible observations. This encourages the encoder to learn meaningful spatial and temporal representations from incomplete video information [135, 125, 40].

- **Vector-Quantized Auto-Encoding (VQ-AE):** In this approach, the encoder first extracts continuous latent features from the video input. These features are then quantized by mapping them to the nearest vector in a finite set of learned codebook embeddings/vectors. The decoder reconstructs the original video from these vector, encouraging the model to learn compact and semantically meaningful representations, due to the limited set of learned embeddings. The underlying principle is similar to language modelling, where a finite vocabulary of discrete tokens is used to represent words, syllables or letters. Analogously, VQ-AE learns a finite set of discrete visual tokens that capture recurring patterns in video data. These tokens may represent object appearances, motion patterns, textures, actions, or scene structures [19, 131, 152, 147, 106].

`Student-Teacher Distillation Approaches` learn representations by transferring knowledge from a pretrained teacher network to a smaller student model [158, 136, 60]. The student is trained to match the representations or predictions generated by the teacher model. This encourages the student network to learn robust and generalized video representations while often requiring fewer computational resources.

`Joint-Embedding Predictive Architectures (JEPA)` learn representations by predicting high-level latent embeddings instead of reconstructing raw video pixels. Rather than modeling every visual detail, the model predicts semantic representations of future or masked observations in a shared embedding space. This approach improves robustness against the high-dimensionality and noise commonly present in video data while focusing on semantically relevant information [8].

### 5.1.2 Video Prediction Models

Video prediction models (VPMs) aim to model the conditional distribution over future visual observations given past context. The standard formulation is next-frame prediction

$$p(o_{t+1} \mid o_{t-k:t}),$$

while more generally VPMs can be expressed as conditional video generation models

$$p(\tau \mid c),$$

where $\tau$ is a video sequence and $c$ denotes conditioning information such as past observations, actions, robot actions, language instructions, or goal images. Trained on large-scale video data, these models implicitly learn priors over physical dynamics, object interactions, and human behavior, making them relevant for robotic learning and imitation learning.

During training, the model learns to predict future visual outcomes from past context, either in pixel space or in a learned latent space:

**Relevance for Robotics.**

Video prediction models are useful in robotics in several complementary ways:

- `Dynamics Models:` Video prediction models can be extended into action-conditioned dynamics models of the form

$$p(o_{t+1} \mid o_{t-k:t}, a_t),$$

to serve as simulator or planner. In this setting, the model predicts future observations conditioned on actions, which allows model-predictive control by evaluating candidate action sequences in visual or latent space.

- `Policy Learning:` VPMs implicitly learn the distribution of behaviors present in the training data by modeling how visual scenes evolve over time. This allows them to generate multiple plausible future trajectories given a current observation. In a robotics context, these generated trajectories can be interpreted as candidate action outcomes.

  This enables an "observation-as-action" perspective, where control is not performed by directly predicting actions, but by evaluating or selecting among predicted future video rollouts. The resulting policy can be viewed as choosing the trajectory that best aligns with a desired objective (e.g., reaching a goal state or matching expert demonstrations), thereby directly connecting video prediction with imitation learning and planning.

- `Representation Learning:` Learned latent spaces capture semantically meaningful information about objects, motion, and interactions, which can be transferred to downstream imitation learning and reinforcement learning tasks.

- `Reward Learning:` Video prediction models can also be used to define reward functions by measuring how well a robot's predicted future aligns with desired behaviors. For example, a reward signal can be derived from the similarity between generated trajectories and expert demonstrations in latent space, or from the likelihood of a trajectory under a learned video prior. This enables imitation learning without manually engineered reward functions.

**Methods**

Modern video prediction approaches are dominated by diffusion models, autoregressive transformers, and masked transformers. Each of these approaches has unique strengths and limitations, and they can be combined with various conditioning mechanisms to enable controllable video generation for robotics applications.

**Diffusion Models.**

Diffusion-based video prediction models generate future frames through an iterative denoising process. During generation, the model starts from random sampled noise and progressively reconstructs a plausible future video sequence conditioned on previous observations, actions, or other conditioning inputs. The denoising model learns to reverse a forward corruption process in which noise is gradually added to training samples.

Diffusion models are particularly effective at modeling complex continuous distributions and producing high-fidelity video samples. However, they suffer from two main limitations: slow sampling speed due to the iterative denoising process, and difficulty maintaining long-horizon temporal consistency across generated frames.

To improve computational efficiency, many approaches perform diffusion in a learned latent space rather than directly in pixel space. In addition, many methods leverage large-scale image datasets alongside video data during training to improve visual quality and generalization. Below are the key differences between pixel-space and latent-space diffusion models:

- **Pixel-Space Diffusion:** In pixel-space diffusion, noise is added directly to raw video frames during training. The model learns to denoise these noisy frames to reconstruct the original video conditioned on context information such as previous frames, actions, or language instructions. While this approach can capture fine-grained visual details, it is computationally expensive and may struggle with long-horizon consistency due to the high-dimensionality of pixel space.
- **Latent-Space Diffusion:** In latent diffusion models, the video frames are first encoded into a lower-dimensional latent representation using an encoder before noise is added. The diffusion process is then performed in this compressed latent space instead of directly on pixels. The denoised latent representation is subsequently decoded back into video frames using a decoder. Operating in latent space reduces computational cost and removes low-level pixel redundancy while preserving the high-level semantic and temporal structure of the video, leading to improved efficiency and better long-horizon consistency.

**Autoregressive and Masked Transformer Models.**

Rather than generating future frames through iterative denoising like diffusion models, transformer-based approaches model videos as a sequence of discrete tokens, typically obtained from a vector-quantized autoencoder (VQ-AE). The encoder maps the input video into a lower-dimensional continuous latent representation. These latent features are then quantized by replacing each feature vector with the nearest embedding vector from a learned codebook. As a result, the video can be represented as a sequence of discrete visual tokens, analogous to words in natural language processing.

Autoregressive transformers generate future tokens sequentially by predicting the next token conditioned on previously generated tokens. While effective at modeling temporal dependencies, small prediction errors can accumulate over long horizons, leading to temporal drift and reduced generation quality.

Masked transformer models address this limitation by masking subsets of video tokens and predicting multiple missing tokens in parallel instead of generating them strictly one-by-one. The model iteratively refines the masked predictions until a complete video sequence is reconstructed. During inference, the model can be conditioned on a short history of recent camera observations from the robot environment, which are encoded into visible tokens, while the tokens corresponding to future time steps are initially masked. The model then predicts the missing future tokens for the entire horizon in parallel and iteratively refines these predictions until a coherent future video sequence is obtained. The model can also be conditioned on additional inputs such as language instructions, goal images, or robot state information, allowing it to generate future video predictions that are aligned with a desired task or behavior. Compared to autoregressive decoding, masked transformers improve computational efficiency and reduce long-horizon error accumulation.

#### Conditioning mechanisms

VPMs can be conditioned on actions, language, or goal images, enabling structured control over generated futures. In robotics, such conditioning effectively turns video prediction models into controllable world models, bridging perception, planning, and imitation learning. With such a world model, a robot can perform counterfactual reasoning over actions, simulate future outcomes of different control strategies, and evaluate how well predicted trajectories align with expert demonstrations or task objectives.

### 5.1.3 Video-to-Text Models.

Video-to-text models are multimodal systems that map video inputs to natural language outputs:

$$p(text \mid \tau),$$

where $\tau$ is a video sequence and *text* is a natural language description. These models are trained on large-scale video-text datasets and learn to extract semantic information from videos while generating coherent language outputs. Video-to-text models can be used to generate natural language descriptions of video content, for video question answering and for video summarization.

#### Relevance for Robotics.

Their usefulness in robotics can be summarized in the following aspects:

- **High-quality representations:** By learning to generate natural language descriptions of video content, video-to-text models learn to extract high-level semantic representations that capture object properties, actions, interactions, and scene context. These representations can be transferred to downstream robotic tasks, improving generalization and task understanding compared to video-only or image-to-text models. The learned representations can capture both high-level semantic information and temporal dynamics, which are critical for understanding complex robotic tasks and environments.
- **Grounded reasoning and planning:** Large language models (LLMs) have demonstrated strong capabilities as high-level planners in robotics. However, their lack of direct grounding in physical perception limits their reliability in real-world environments.

Video-to-text models address this limitation by conditioning reasoning directly on rich visual-temporal input. This enables:

- improved grounding of symbolic reasoning in observed physical states
- better modeling of object interactions and environment dynamics
- more consistent closed-loop reasoning, where predictions are continuously informed by perceptual feedback

- **Annotating robot data:** Video-to-text models can be used to generate high-quality language annotations for robot datasets. By automatically describing video demonstrations of robotic tasks, these models can create rich textual annotations that capture task-relevant information such as object properties, actions, goals or step-by-step task descriptions.
- **Rewards and Feedback:** Video-to-text models can be used to provide reward signals or value estimates for reinforcement learning and imitation learning systems. By framing reward learning as a visual-question answering problem, the model can evaluate a robot's behavior directly from video observations. For example, the model may answer questions such as "Did the robot successfully grasp the object?" or "Did the trajectory achieve the desired goal?". This formulation is typically goal-oriented and explicitly tied to predefined task semantics. The predicted responses can then be transformed into reward signals that guide policy optimization.

  In addition, video-to-text models can be integrated into reinforcement learning from AI feedback (RLAIF) frameworks, where the model evaluates the quality of predicted or executed behaviors relative to task objectives or expert demonstrations. The model compares different behaviors—either whole trajectories or segments of trajectories—and provides judgments such as which behavior is more successful, safe, or aligned with the desired objective. These ranking-based signals are then distilled into a reward model or value function.

**Methods**

`Video-to-Text Methods` Existing approaches can be broadly categorized into compositional pipelines, LLM-based architectures with adaptors, and natively multi-modal sequence models.

- **Compositional Approaches:** decompose the video-to-text problem into smaller sub-tasks. Typically, an Image-Language Model is first used to process individual video frames by answering frame-level questions. The resulting intermediate descriptions are then aggregated using a Large Language Model (LLM), which generates a global video summary. While effective in practice, these approaches do not involve end-to-end video training, and therefore often fail to learn rich temporal video representations.
- **Leveraging Pretrained LLMs via Adaptors and Finetuning:** A common strategy is to combine a pretrained video encoder with a pretrained LLM. Since LLMs

operate on token sequences, an adaptor module is introduced to project video features into the LLM input space. With the adaptor, the video encoder and LLM can be trained end-to-end on video-text data, allowing the model to learn to extract video representations that are aligned with language generation.

- **Natively Multi-Modal Models:** More recent approaches integrate video and text into a unified autoregressive framework, instead of combining separate models.

  In natively multi-modal sequence models, video-to-text generation is formulated as an interleaved sequence modeling problem over visual and textual tokens. Video inputs are first tokenized into discrete visual tokens, which are then combined with text tokens into a unified sequence.

  The model is trained autoregressively using next-token prediction over this joint sequence, i.e.,

  $$p(x_t \mid x_{<t}),$$

  where tokens may correspond to either visual or textual elements. This formulation enables a unified treatment of different modalities within a single transformer architecture.

  As a result, such any-to-text or any-to-any models allow video, text, and potentially other modalities to interact within the same latent space. This leads to a tighter coupling between vision and language representations and moves towards more general multi-modal foundation models capable of flexible cross-modal generation and reasoning.

### 5.1.4 Multi-Modal Models

Multi-modal models jointly learn from multiple data modalities, which can include text, images, videos, robot observations, and robot actions. Unlike specialized vision-language models, these approaches aim to learn a unified sequence model that can process heterogeneous modalities within a shared architecture, that can connect visual perception, language understanding, and robot control.

Most multi-modal models are trained using autoregressive next-token prediction objectives over interleaved sequences of tokens from different modalities. Depending on the model, tokens may represent text, image patches, video frames, proprioceptive robot observations, or robot actions. The objective is to predict the next token in the sequence conditioned on the previous context:

$$p(x_t \mid x_{<t}).$$

where tokens may correspond to different data modalities or a combination or concatenation of them. This training paradigm allows the model to learn cross-modal relationships between perception, language, and control.

**Relevance for Robotics.**

- **High-quality representations:** By training on robot data together with web-scale video, image, and text data, these models learn a shared latent space that bridges perception, language, and control. This enables positive transfer and can serve as general-purpose representation backbones for robotic learning.
- **Knowledge Modality initialization:** The supervised multi-modal model can serve as a strong inital Knowledge Modality, such as Policies, Reward Models, Value Functions, or Dynamics Models. For Example, RL can fine-tune an inital Policy using task-specific rewards. This two-stage approach (supervised pre-training + RL finetuning) has been shown to improve sample efficiency and final performance compared to RL from scratch.
- **Annotating robot data:** Multi-modal models can also be used to generate annotations for robot datasets.

**Relation to Vision-Language-Action (VLA) Models.**

Multi-modal models that explicitly include robot actions are often called Vision-Language-Action (VLA) models. However, the distinction between both categories is not always strict. VLA models typically focus on direct policy inference, where robot actions are predicted from visual observations and language instructions, while the multi-modal models discussed here are general-purpose token predictors that model arbitrary modality sequences. In practice, the same underlying architecture can often be used both as a general multi-modal foundation model and as a robotic control policy.

### 5.1.5 Challenges and Future Directions

Despite recent progress, video representation learning still faces several limitations:

- Current models may **hallucinate** or fail to capture fine-grained spatial relationships, precise spatio-temporal dynamics, and long-term temporal dependencies.
- Another key bottleneck is the **limited quality and scale of low-level annotated video data**, particularly for tasks requiring detailed fine-grained understanding.
- Additionally, processing **high-dimensional and long video sequences** remains computationally expensive, motivating more efficient model architectures.

Further directions include reinforcement learning-based finetuning to reduce hallucinations, the integration of 3D information, like depth to improve physical realism, and improved single-step conditioning mechanisms for fine-grained video prediction tasks.

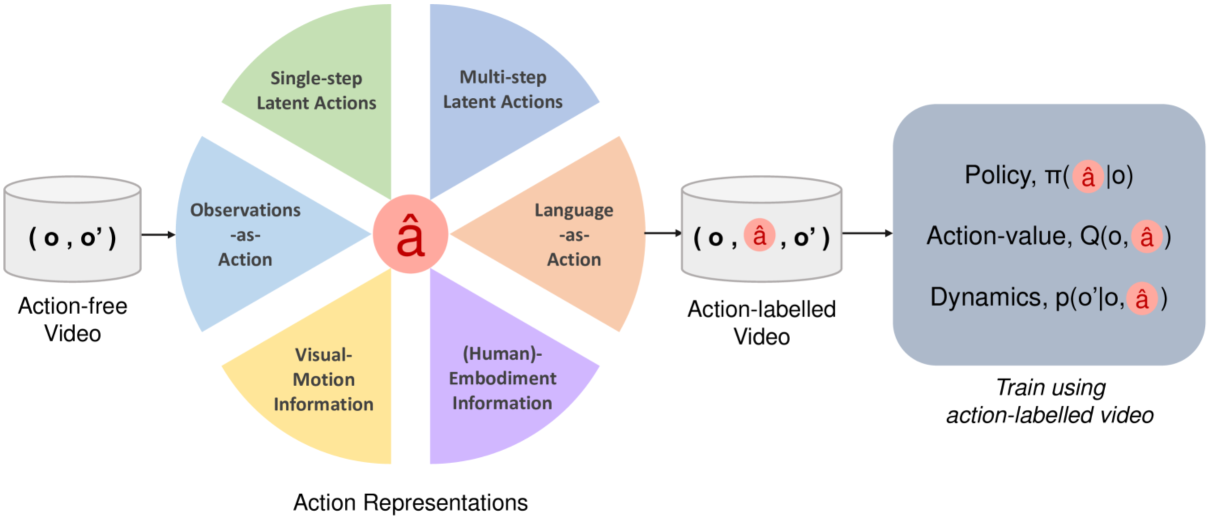



**Figure 5.2:** To label action-free videos, an Alternative Action $\hat{a}$ can be derived from video data instead of the missing action $a$ [73].

## 5.2 Video-to-Robot Transfer: Addressing Missing Action Labels and Distribution Shift

The main challenges for transferring video-extracted knowledge to robotics are:

- (i) **Missing action labels**: Most video datasets do not contain explicit robot action annotations, which are required for learning robot control policies [73].
- (ii) **Distribution shift**: There exists a significant domain gap between human or internet videos and robot execution. This includes differences in visual appearance, embodiment, dynamics, and behavior patterns, which can severely degrade the generalization of models trained on video data when applied to robotic tasks [73].

In many real-world video datasets, explicit robot action labels (e.g., joint velocities or torques) are not available. Raw video data provides only state transitions pairs $(o_t, o_{t+1})$, without access to the corresponding actions $a_t$. This makes it difficult to directly apply Behavior Cloning or reinforcement learning methods, which typically require triplets of the form $(o_t, a_t, o_{t+1})$. Recovering such low-level actions directly from video is a challenging problem due to ambiguity and the lack of physical interaction information [73]. To overcome this limitation, recent approaches introduce an Alternative Action Representation $\hat{a} \in \hat{A}$, which can be used to relabel action-free videos and enable video-to-robot transfer despite missing action labels and distribution shift, see Figure 5.2.

Formally, this induces a relabeling of video transitions as:

$$(o_t, o_{t+1}) \;\rightarrow\; (o_t, \hat{a}_t, o_{t+1}), \quad \hat{a}_t \in \hat{A}.$$

### What is an Alternative Action Representation?

Formally, an alternative action representation is defined as

$$\hat{a} \in \hat{A},$$

where $\hat{A}$ denotes an Alternative (latent or abstract) Action Representation Space. Instead of modeling actions as raw robot control signals, actions are encoded in a semantic feature space that captures high-level behavior such as grasping, moving, or pushing. In other words, $\hat{A}$ captures what actions *mean*, rather than how it is executed at the motor level [73]. The key idea is that $\hat{a}$ serves as a proxy for the missing action labels, allowing the model to learn from video data without requiring explicit action annotations. Later, the $\hat{a}$ can be decoded or translated into the robot's actual action space if a mapping $f : A \rightarrow \hat{A}$ exist [133, 138, 29]. Otherwise, they still provide valuable structure [12, 105]. However it is also valid to define a $\hat{A}$ that requires labelled video (e.g., language captions) [73].

Example: Consider a video of a human performing a pick-and-place task. The raw video data provides visual observations of the human's hand and the objects being manipulated, but does not include explicit robot action labels such as joint angles or end-effector velocities.

- The **Video Encoder** extracts visual features (what is being done), see section 5.1.1.
- The **Action Embedding Space** $\hat{A}$ captures *abstract actions* such as *reach*, *grasp*, *move up*.
- The **Policy** $\pi_{alt}(\hat{a}_t|o_t, c), \quad \hat{a}_t \in \hat{A}$ predicts the next action *in this latent action space* based on current visual observations and probably some additional context information $c$, see section 6.2.4.
- Later, a **Decoder** $\pi(a_t \mid \hat{a}_t, o_t)$ or **Alignment Model** $a_t = f_{align}(\hat{a}_t)$ maps these latent actions to the robot's real motor commands, see section 6.2.4.

### Why is $\hat{A}$ useful?

Working in the alternative action space $\hat{A}$ offers several advantages.

- `Bridging the domain gap:` $\hat{A}$ bridges the gap between human video data and robot control by focusing on action semantics rather than platform-specific motor commands [73].
- `Learning from weak labels:` $\hat{A}$ enables learning from raw or weakly labeled videos, since inferring high-level actions from video is easier than recovering low-level motor commands [73].
- `Transferable Representations across embodiments:` $\hat{A}$ supports transfer across embodiments, because the same semantic action (e.g., "open drawer") can be instantiated by different robots with different kinematics [73].

- `Structured learning:` A well-structured representation $\hat{A}$ that is minimal, disentangled, and semantically consistent across the state space encodes meaningful action information. This makes it easier to learn models that are predictive of both future observations (i.e., learning $p_{\text{alt}}(o_{t+1} \mid \hat{a}_t, o_t)$ instead of $p(o_{t+1} \mid o_t)$) and robot actions (i.e., learning $\pi(a_t \mid \hat{a}_t, o_t)$ instead of $\pi(a_t \mid o_t)$), leading to more data-efficient and generalizable policies, dynamics models, and value functions [73, 101, 19]. The choice of abstraction level for $\hat{A}$ also plays a crucial role:
  - **Higher-level or longer-horizon representations** $\hat{A}$ (such as language-based actions) can benefit performance in complex, long-horizon tasks by providing broader context and intent. This is because they capture more abstract and semantically meaningful information that can guide decision-making over longer time scales, which is particularly beneficial for tasks that require planning and reasoning about future consequences [73].
  - **Lower-level or shorter-horizon representations** $\hat{A}$, which retain more fine-grained information, may be more effective for accurately reconstructing or decoding the original robot actions $a_t$ from the alternative actions $\hat{a}_t$ [73].

Once $\hat{A}$ is defined, the same representation can be used for example for:

- policies $\pi_{\hat{A}}(\hat{a}_t \mid o_t)$ [105],
- dynamics models $p_{\hat{A}}(o_{t+1} \mid o_t, \hat{a}_t)$ [19] or
- state-action value functions $Q_{\hat{A}}(o_t, \hat{a}_t)$ [12],

which further encourages consistent and transferable action abstractions.

### 5.2.1 Categories of Alternative Action Representations

Their exist different approaches to derive alternative action representations $\hat{a} \in \hat{A}$ from video data. These methods differ in how actions are represented and which information is extracted from visual observations [73].

**Single-step Latent Actions:**

These approaches learn latent actions that describe the transition from one observation to the next [73]:

$$\hat{a}_t : o_t \to o_{t+1}$$

**Methods:**

- Forward Dynamics Models (FDM) $p(o_{t+1} \mid \hat{a}_t, o_t)$, which predict the future observation from the current observation and latent action [19, 31, 105, 151, 101].
- Inverse Dynamics Models (IDM) $p^{-1}(\hat{a}_t \mid o_t, o_{t+1})$, which infer the latent action responsible for a transition [105, 19].

**Limitations:**

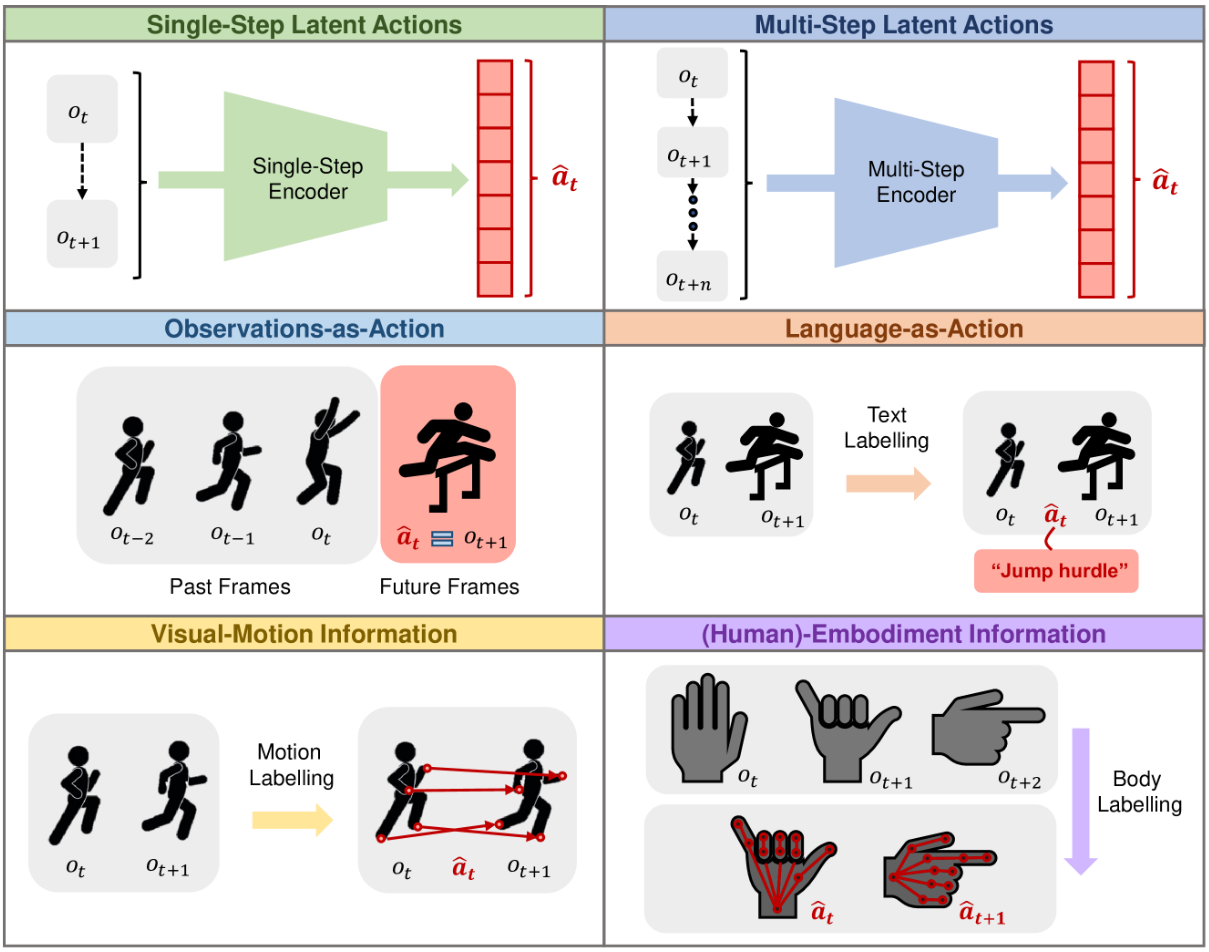



**Figure 5.3:** Categories of Alternative Action Representations that can be learned or obtained from video data [73].

- Often capture all visual changes, not only agent actions [73].
- May omit important non-visual information such as forces or contact dynamics [73].

**Multi-step Latent Actions:**

These methods encode behaviors across multiple time-steps:

$$\hat{a}_t : o_t \rightarrow o_{t+k}$$

Instead of modeling individual actions, they capture higher-level behavioral patterns or latent plans [73]. The key idea is to learn a compact latent representation of a *skill* or *sub-policy* that summarizes an entire sequence of actions. These representations are more abstract, goal orientated and can encode longer-horizon information, which may be beneficial for complex tasks [100, 25, 69].

For example these "skill embeddings" can capture structures such as "reach the object", "move the arm to the target location", or "grasp and lift", which can later be decoded into a full action sequence.

**Methods:**

- Variational sequence models: These methods use variational autoencoders (VAE) to encode motion trajectories or sequences of observations into a latent variable $z$ that captures the underlying behavior. The VAE is trained to reconstruct the original trajectory from the latent representation [133].
- Video-language contrastive learning aligns video segments with natural language descriptions of behaviors [34, 62, 115].
- Supervised contrastive learning on action-labeled videos pulls videos with the same or similar action labels together in latent space, while pushing videos with different action labels apart [21, 22].
- Self-supervised clustering approaches group similar motion patterns or trajectories into clusters without requiring labels, with each cluster representing a reusable skill or behavior primitive [145].
- Further approaches are presented in the works of Tomar et al. [124], Pertsch et al. [90], and Cai et al. [20].

**Limitations:**

- Often produce abstract and long-horizon representations that may lose fine-grained control details and be harder to align with low-level robot actions [52, 61].

**Observations-as-Actions:**

These approaches directly use (encoded) future observations as action representations:

$$\hat{a}_t = o_{t+k}.$$

Future observations provide information about the intended behavior or goal state [73].

**Methods:**

- `Next-observation-as-action` uses the next observation $o_{t+1}$ as the $\hat{a}_t$ label and provides information regarding the action that should be taken [29, 121, 43].
- `Observations-as-subgoals` uses future observations as intermediate goals to guide action selection [13, 85, 12].
  - `Fixed time horizon` uses observations that are fixed $k$ time steps in advance [13, 29].
  - `Random time horizon` uses observations that are a randomly sampled number of time steps in advance [12].
  - `Key-frame selection` identify critical states in a trajectory that can serve as action representations [91, 64], see table 5.1.

| Time Step | Frame Description | Significance |
|---|---|---|
| $t_0$ | Arm over table | Start |
| $t_1$ | Arm approaches cup | Intermediate state |
| $t_2$ | **Hand touches cup** | ✓ Key-frame |
| $t_3$ | Cup is lifted | Intermediate state |
| $t_4$ | **Cup over target** | ✓ Key-frame |
| $t_5$ | Cup is placed | End |

**Table 5.1:** Example Key-Frames in a Pick-and-Place Sequence

**Limitations:**

- Future observations may not uniquely determine the underlying action [57].
- High-dimensional visual observations can increase learning complexity [57, 68].

**Language-as-Actions:**

Natural language descriptions can be used as high-level action representations for example by captioning video frames with language instructions that describe the observed behavior [9, 110, 140]:

$$\hat{a}_t = \texttt{"pick up the cube"}.$$

Language-based actions provide semantic supervision and enable integration with large language models [30], which can facilitate more complex reasoning and planning.

**Methods:**

- Language-captioned video datasets [42]
- Process and enhance standard captions to generate more detailed action descriptions, for example by using LLMs or VLMs [78].
- Automated or semi-automated video captioning [78]
- Video-to-Text Models [73]
- Vision Language Models (VLMs) and variants of them [73]

**Limitations:**

- Language is often coarse and omits low-level control details [73].
- Requires annotated or captioned video data [73].

**Visual Motion Information:**

These approaches define actions using visual motion cues extracted from video sequences [73]. The underlying assumption is that temporal changes in visual observations (motion patterns) contain implicit information about the actions responsible for observed state transitions. For example, the direction, magnitude, and trajectory of movement can provide information about the intended behavior, goal, or interaction dynamics [150, 123].

**Methods:**

- `2D or 3D point trajectories` of tracked objects or hands [138, 154] via off-the-shelf point trackers [51]
- `Optical flow prediction`, which captures how image regions move between consecutive video frames [56, 143]
- `Visual action arrows` that represent movement direction and magnitude directly in the image [82]
- `Structure-from-motion` approaches, which reconstruct 3D scene geometry and camera motion from video in order to recover information about underlying physical interactions [134, 155, 148]

**Limitations:**

- Motion estimates can be noisy or ambiguous [14].
- Visual motion does not always correspond directly to robot actions [14].

**(Human)-Embodiment Information:**

These approaches explicit models human body structure and kinematics [73, 11, 109, 5, 94, 95]. In contrast to generic motion-based methods, which focus on pixel or object movement, these approaches aim to represent the *agent performing the action* using structured human pose information. Embodiment-specific information such as hand poses, affordances, or trajectories can provide useful cues about the intended actions and interactions in the video, which can be transferred to robotic embodiments [99, 107]. For example, an alternative action representation can be defined as:

$$\hat{a}_t = \text{hand pose at next time-step } t+1,$$

which encodes the intended end-effector movement and can be associated with actions such as reaching, grasping, or manipulating objects. Animal embodiment could also be usefull [87].

**Methods:**

- Human pose estimation (full-body kinematics) [11]
- Hand pose estimation and hand trajectory tracking [99, 107]
- Affordance prediction (e.g., graspable regions, interaction points, grasp trajectories) [5, 11]
- Object- and hand-mask conditioned representations, which focus on the regions of the video that are relevant for action understanding and ignore irrelevant background information [11]

**Limitations:**

- Human embodiment is structurally different from robotic embodiments, limiting direct transferability.
- Accurate pose and hand tracking or estimation can be unreliable in cluttered or occluded scenes.
- Focuses on human-centric actions, which may not generalize to non-human interaction styles.

**Inverse Dynamics Model (IDM) pseudo-action Labels:**

These methods train an inverse dynamics model on robot data and use it to infer pseudo-action labels for action-free (human) videos [7, 126, 104], which can then be used for learning policies [73]:

$$p^{-1}(a_t \mid o_t, o_{t+1}).$$

**Limitations:**

- Require action-labeled robot data for model training [7, 126, 104].
- Small domain shift between demonstration videos and robot execution [7, 104, 53].

## 5.3 LfV and Reinforcement Learning

# Chapter 6

# Robot Learning Fundamentals

## 6.1 Robot Learning - Learning Objectives

Learning robot manipulation skills needs a comprehensive visual pipeline and objectives such as learning meaningful representations of the environment, understanding object affordances and human-object interactions, recognizing human actions and activities, and modeling 3D hand movements for fine-grained manipulation. These objectives are crucial for enabling robots to perform complex tasks in dynamic environments.

### 6.1.1 Representation Learning

**Goal:** Representation Learning aims to extract meaningful and task-relevant spatio-temporal features from large amounts of unlabeled visual data. In robotics and Imitation Learning, the goal is to learn representations that enable agents to interpret dynamic scenes, understand interactions, and generalize across varying environments and embodiments.

A key challenge is that robotic observations are high-dimensional and subject to variations such as viewpoint changes, lighting conditions, cluttered backgrounds, and partial occlusions. Therefore, learned representations should be invariant to irrelevant visual changes while preserving information relevant for control and decision making.

Modern representation learning methods focus on several key aspects:

- **Spatial Understanding:** Capturing objects, scene layouts, semantic structures or patterns, and relationships between entities.
- **Temporal Modeling:** Understanding motion dynamics, temporal coherence, and action progression across video frames.
- **Spatio-Temporal Modeling:** Extracting features within a frame (spatial) and tracking changes across frames (temporal coherence and dynamics)

- **Geometry-Aware Representations:** Understanding of three-dimensional scene geometry. Geometry-aware representations enable the agent to infer depth, 3D structure, and spatial consistency from monocular or multi-view visual observations. This capability is particularly relevant for robotic manipulation tasks, where accurate spatial reasoning is a prerequisite for grasp planning, obstacle avoidance, and trajectory optimization
- **Robustness and Invariance:** Learning consistent representations across multiple views, lighting conditions, and environmental variations.

**Understand the importance of Representation Learning:**

1. **Spatial Feature Extraction (The "What" and "Where")**

   *Definition:* Spatial feature extraction focuses on what is happening within a single video frame. It involves identifying patterns, objects, and relationships based on pixel data in a 2D space.

   *Analogy:* If you pause a video, the spatial features are everything you can identify in that still image—the location of the object, the color of the table, and the shape of the hand.

   *Robot Relevance:* Answers the question: "What is the current state of the environment?" (e.g., The red block is on the left.)

2. **Temporal Feature Extraction (The "When" and "How")**

   *Definition:* Temporal feature extraction focuses on the relationships and changes that occur across multiple frames over time (the sequence).

   *Analogy:* This is the motion and dynamics of the video—understanding that a hand is moving toward a block, or that the block is rotating.

   *Robot Relevance:* Answers the question: "What is the action and how is the scene changing?" (e.g., The hand is picking up the block.)

3. **Incorporate geometry and structure**

   *Definition:* When your spatial and temporal methods incorporating geometry and structure, it refers to features that move beyond the 2D pixel domain (spatial) and the sequence domain (temporal) to model the 3D physical world.

   *Analogy:* This involves estimating properties like depth (how far away objects are) and camera motion/pose (the 3D position of the camera).

   *Robot Relevance:* For robotic manipulation, having a 3D and geometry understanding is essential, as the robot must operate in a 3D space, not just a 2D image.

Video-centric representation learning is particularly important in robotics because it explicitly models temporal dynamics and multi-view consistency. These learned representations form the backbone of vision-based perception and control systems, enabling the transfer of high-level behavioral information from human videos to robotic policies. As a result, strong representations can significantly improve generalization, robustness, and data efficiency in imitation learning systems or facilitate more robust sim-to-real transfer.

**Methods.**

Video-centric methods (see Section 5.1) are especially relevant for Robot Representation Learning, as they explicitly model temporal dynamics and multi-view consistency.

Recent approaches in representation learning leverage large-scale, unlabelled video data and self-supervised objectives to extract spatio-temporal features from raw observations [35, 36]. Works have often combined multiple learning objectives [188, 170, 125] for Representation Learning. Learning objectives for Representation Learning can include:

- Frame-level objectives such as **masked-autoencoding (MAE)** [213]
- **contrastive learning** [52, 120].
- others [328]
- **Time-contrastive learning objectives** [171, 170, 188]:
- **video prediction objectives** [289, 95, 116];
- **temporal-difference learning** [27];
- **predicting affordances from videos** [15];
- **predicting latent actions** [227];
- objectives that leverage **language labels**, such as **image-language contrastive learning** [170], **video-language alignment** [188] and **video captioning** [125].

**Combining datasets** to enhance data diversity is beneficial [120] [174, 61, 213]. Datasets may include the following types of videos:

- Datasets often leverage ego-centric human/navigation video [188, 170, 125, 174]
- or third-person video of human interaction [213, 239, 181].
- Others have leveraged videos of robots [61],
- or first-person videos of humans operating robot-like grippers [237].

### 6.1.2 Object Affordances & Human-Object Interaction

Object affordances stands for the actionable properties of objects, namely the interactions and manipulations that an object enables or invites. In other words it describes what you can do with an object for example:

- a handle affords *grasping* or *holding*

- a button affords *pressing*
- a drawer handle affords *pulling*

In robotics and imitation learning, affordance understanding aims to infer how humans interact with objects and how these interactions can be transferred to robotic manipulation policies.

The central goal is to understand the functional relationships between humans, objects, and actions by observing human manipulation in videos. Unlike purely appearance-based object recognition, affordance learning focuses on *how objects are used* and interacted with over time.

Modern affordance learning methods typically address several important aspects:

- **Affordance Grounding:** Affordance grounding refers to the high-level semantic understanding of object functionality. The goal is to identify which actions an object enables and relate them to meaningful interactions.

  *Example:* Scissors afford cutting, while a mug affords holding liquids.

- **Interaction Region Localization:** Interaction region localization focuses on identifying the specific object regions that are relevant for manipulation and interaction. Rather than only understanding what actions an object affords, the goal is to determine where these interactions should physically occur, such as grasp points, handles, buttons, or contact surfaces.

  *Example:* For scissors, the handle region is associated with grasping, while the blades correspond to the cutting interaction. Similarly, for a mug, the handle represents the primary grasping region.

- **Spatio-Temporal Affordances:** Capturing how affordances evolve spatially over time during manipulation sequences and task execution. This includes interaction dynamics, hand-object contact patterns, grasp types, trajectories, and sequential manipulation behavior.

  *Example:* Cutting paper with scissors requires coordinated hand motion, repeated opening and closing actions, and continuous contact between the blades and the paper. Similarly the affordance of a door handle changes from "graspable" to "turnable" and then to "pulling" during interaction, reflecting the temporal dynamics of the manipulation process.

- **3D and Geometry-Aware Understanding:** Incorporating depth information, RGB-D observations, point clouds, and multi-view inputs to the affordance model can improve spatial reasoning and manipulation robustness.

  *Example:* Estimating the 3D orientation of scissors and their distance to the paper enables more precise and physically consistent manipulation.

**Methods.**

Recent approaches increasingly leverage large-scale human activity videos and self-supervised learning to infer affordances from natural human demonstrations [48]. Hand-centric representations are particularly important, as hand motion and contact points provide strong cues about object functionality and intended actions [49]. Several methods therefore use hand localization, hand-state estimation, or hand-object contact prediction to improve affordance grounding and interaction understanding.

The usage of transformer architectures and the integration of RGB-D sensing and multi-modal inputs has further improved the ability to jointly reason about objects, actions, spatial-temporal dynamics, and scene context. In addition, 3D point cloud representations and multi-view observations help reduce uncertainty in manipulation planning and improve robustness in real-world robotic environments.

Affordance learning plays a critical role in robotic imitation learning by bridging the gap between observing human demonstrations and executing robotic actions. By learning functional object understanding rather than purely visual appearance, robots can better generalize manipulation skills across environments, viewpoints, and object instances. This enables more robust action grounding for tasks such as grasping, opening, placing, tool use, and long-horizon manipulation.

### 6.1.3 Human Action & Activity Understanding

**Goal:** Human Action and Activity Understanding aims to recognize what humans are doing in videos, which is crucial for understanding object affordances. It allows robots to interpret fine-grained manipulation skills, replicate task structure, timing, and sequencing from human demonstrations. The goal is to learn a hierarchical understanding of human actions, from low-level movements to high-level activities, by analyzing visual cues such as hand trajectories, object interactions, and scene context.

Modern methods therefore focus on several essential aspects:

- **Temporal Action Identification and Segmentation:** Modeling the temporal structure of tasks by decomposing complex activities into meaningful sub-actions or atomic interaction steps, allowing robots to understand the sequential structure of a task [68, 69].
- **Fine-Grained Action Recognition:** Distinguishing visually similar manipulation differences (e.g., grasping vs. pinching, rotating vs. twisting) by jointly leveraging appearance and motion cues.
- **Motion Flow and Interaction Region Understanding:** Predicting keypoints, interaction regions, and motion trajectories from demonstrations to capture where and how human hands interact with objects over time [68].
- **Hierarchical Task Intention Modeling:** Learning relationships between fine-grained low-level actions and high-level task goals, Allows robots to understand what humans are doing and replicate task structure, timing, and sequencing.

**Methods.**

Recent advances include unsupervised and weakly supervised methods that reduce the need for dense annotations. In [69], the method models complex actions as sequences of latent sub-activities by alternating between a discriminative step, which separates meaningful temporal segments, and a generative step, which reconstructs or explains the observed sequence. An explicit background model is used to suppress non-informative regions (e.g. scene clutter), allowing the system to focus on action-relevant frames under weak supervision. IRMT-Net [68] directly learns interaction structure from demonstration videos by jointly predicting spatial interaction regions and corresponding motion trajectories, effectively coupling "where interaction happens" with "how it evolves over time," which reduces dependence on manual annotations and improves generalization across settings.

### 6.1.4 3D Hand Modeling

**Goal:** 3D hand modeling aims to estimate accurate 3D pose, shape, and contact of the human hand from video, to directly translate human hand movements into robot gripper commands.

By reconstructing a full 3D mesh of the hand, the system can understand fine-grained manipulation details—such as finger placement, grasp type, and hand-object contact—and retarget these to robot end-effectors. 3D Hand Modeling helps to bridge the embodiment gap between human hands and robot grippers [71, 72, 75].

Key challenges include handling occlusions (e.g., self-occlusion of fingers), recovering realistic hand-object contact, and mapping human joint angles to robot kinematics. Modern methods therefore focus on several essential aspects:

- **Parametric Hand Models:** Low-dimensional, realistic representations of the human hand (e.g., MANO [75], built on SMPL [76]) that enable 3D mesh reconstruction from monocular images [73, 74]. These models capture hand pose, shape, and articulation while being computationally efficient for real-time applications.
- **Occlusion Handling:** Robust estimation of finger positions and hand shape despite self-occlusions or object-induced occlusions during manipulation.
- **Grasp Realism and Hand-Object Contact:** Enforcing consistency between hand contact points and object surfaces, often through self-supervised objectives that improve physical plausibility [60].
- **Motion Retargeting:** Mapping human hand joint angles and poses to robot gripper or anthropomorphic hand commands, enabling transfer of dexterous skills across embodiments [11, 71].

**Methods.**

Advances in motion capture have further improved fidelity. FrankMocap enables fast monocular 3D hand and body pose estimation using SMPL-X [73, 74], whereas MANO provides a parametric hand model widely used in robotics and graphics [75, 76]. Complementary work focuses on grasp realism, enforcing consistency between hand-object contact points through self-supervised objectives, improving flexibility and accuracy even during testing [60].

Collectively, these methods reduce embodiment differences and establish robust hand modeling pipelines, laying the foundation for high-fidelity manipulation learning from human demonstrations [71, 72, 75].

## 6.2 Robot Learning - Approaches

### 6.2.1 Foundational Perception and Representation Methods

The foundation of video-based robot learning relies on the ability to extract task relevant information from visual data and translate it into robot-compatible representations. Robust perceptual and representational foundations are crucial for bridging the gap between human demonstrations and robot execution and enables agents to understand what matters in a video and how to replicate it [33].

This section is divided into:

1. *Feature Extraction Methods* which focus on extracting meaningful structured spatio-temporal features from video data, including spatial understanding, temporal modeling, and geometry-aware representations [33].
2. *Domain Bridging Approaches* which aim to reduce the gap between different domains, such as sim-to-real or human-to-robot transfer, by translating visual or contextual information from videos into robot-compatible representations [33].

**Feature Extraction Methods**

Feature extraction methods aim to obtain task-relevant information from visual demonstrations, ranging from holistic scene understanding to fine-grained motion structure. In robotic imitation learning, these approaches provide the perceptual foundation for mapping visual observations to robot actions [33].

- **CNN-based Feature Extraction:** Convolutional Neural Networks (CNNs) excel at spatial feature extraction and learn spatio-temporal representations directly from raw pixels [96]. These approaches are widely used for detecting objects, segmenting interaction regions, and modeling hand-object relationships from demonstrations [58, 156, 157]. CNN-based pipelines form the foundation of many Robot Learning systems and support the direct mapping of raw visual observations to robot actions [33].

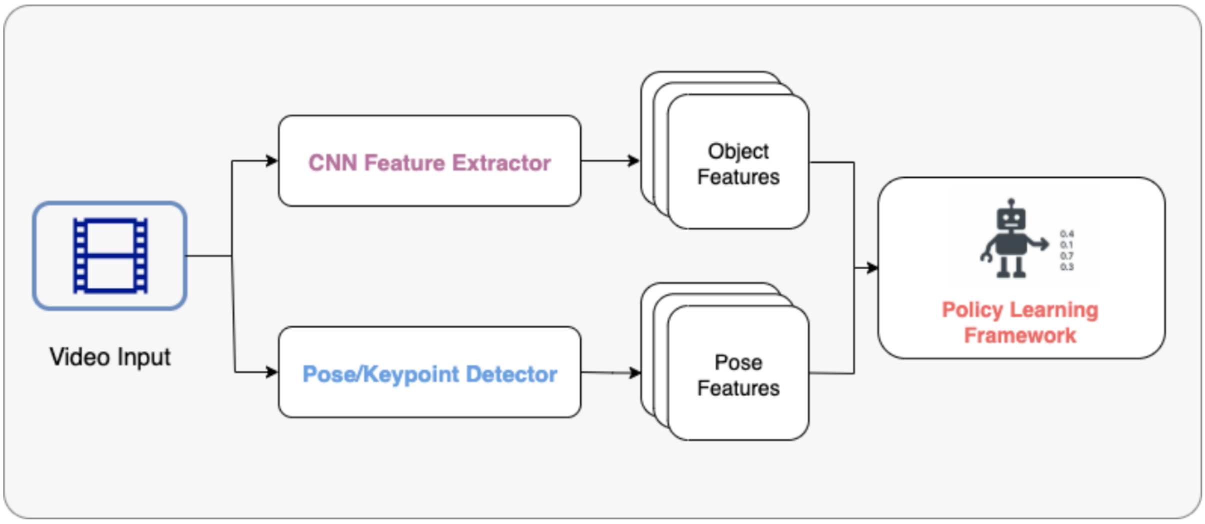



**Figure 6.1:** Feature Extraction Methods [33].

- **Pose and Keypoint Detection:** In contrast, pose and keypoint-based methods focus on extracting structured skeletal motion representations such as body joints, hand poses, fingertips, or object centers. Instead of analyzing the entire image equally, these methods identify important points of the human body or hand, such as elbows, wrists, fingertips, or object locations, and track how they move over time [33]. This helps the system understand how a person moves, how body parts relate to each other, and how humans interact with objects during a task . Pose-based methods are particularly useful for retargeting human demonstrations to robot kinematics, since the reconstructed joint trajectories can be mapped to robot motions [112, 4]. For example, several approaches reconstruct 3D human poses from monocular videos and transfer the resulting trajectories into robot-compatible action representations for policy learning [112, 27, 163].

These pose-based methods complement CNN-based pipelines—while CNNs extract dense, holistic features, pose and keypoint detection supplies structured, low-dimensional skeletal motion cues that ease embodiment alignment and improve robustness in human-to-robot transfer tasks. Both methods can provide valuable information for downstream policy learning.

Feature extraction methods are generally data-efficient and flexible, making them well-suited for learning from large-scale unlabeled video data and producing interpretable representations. They integrate effectively with reinforcement learning and imitation learning frameworks, particularly for modeling perception-driven control and effective for action parsing and object-centric manipulation tasks. However, these methods still struggle with sim-to-real or human-to-robot embodiment transfer, long-horizon or high-level policy reasoning, and domain shift between human demonstrations and robotic execution [33].

**Domain Bridging via Translation Approaches**

This category addresses the domain shift—the mismatch in human and robot embodiment, visual appearance, and context—between human demonstration videos and robot execution. Translation techniques convert visual and contextual data into robot-compatible representations, such that human demonstrations become more directly usable for robot learning [33].

- **Image Translation:** Image Translation techniques focus on aligning visual appearance between human and robot domains. The main idea is to translate human videos directly into a robot-like visual style or make simulated robot data more realistic. For example, mapping the visual appearance of a human arm to the robot arm or the human hand to the robot gripper. This allows robots to learn from human demonstrations while observing them in a visually consistent format [33].

  A common approach is based on GANs (Generative Adversarial Networks), which consist of two competing models: a generator that produces the translated robot images and a discriminator that tries to distinguish between real and generated robot images. The goal is for the generator to generate robot images that look realistic enough that the discriminator cannot distinguish them from real ones. Through this competition, the generator learns to produce increasingly realistic images that match the target domain [45, 161, 66]. CycleGAN based approaches extend this idea by adding a cycle-consistency constraint, meaning that translating an image from domain A to B and back to A should reconstruct the original image [162]. This allows learning translation between human and robot domains without requiring paired examples.

  Based on these principles, early GAN-based methods [45, 161, 66] improved the realism of training data and supported better generalization to robot manipulation tasks [32]. More advanced approaches use conditional GANs to predict intermediate visual goals from demonstrations, which are then executed by a separate low-level controller, effectively separating perception from control [108]. Other methods, such as AVID [113], apply CycleGAN to directly translate human videos into robot-like images, using the resulting frames as reward signals for reinforcement learning. More recent approaches extend this idea with transformer-based architectures, bidirectional translation, meta-learning and RL, enabling fast adaptation to new tasks with minimal demonstrations [26, 59].

- **Context Translation:** Context translation methods extend beyond visual appearance alignment and focus on transferring skills across varying tasks, environments, viewpoints, and object configurations [65]. The main goal is to enable robots to generalize learned behaviors to new contexts by capturing the underlying structure of a task or the relationships between different elements in the environment rather than relying on specific visual conditions. In short terms, the idea is to learn “what is being done” rather than “how it looks” in a particular scene.

Early approaches learn from paired demonstrations collected in different environments, where the same action is shown under varying contexts. These models learn to translate how a skill appears across settings, enabling robots to reproduce the same behavior in novel environments through reinforcement learning [67]. Later methods improve robustness by learning context-agnostic task representations that are combined with multi-modal inverse dynamics models $p^{-1}(\hat{a}_t \mid o_t, c_t, o_{t+1}, c_{t+1})$, by integrating RGB ($o$) and depth (represented as conditional modality $c$) information as the input. These systems achieve more stable action prediction across changes in viewpoint and object layout [149].

More recent approaches focus on scalability through unsupervised learning. The Learning by Watching (LbW) framework [141] combines video translation models with unsupervised keypoint extraction to map human demonstrations into robot domains without task-specific supervision. The resulting structured keypoint representations are then used as inputs for reinforcement learning, enabling imitation under varying contextual conditions while reducing the need for expert-labeled data.

**Example**

Learning goal: The Robot should learn to bake a cake from a YouTube video

1. **Feature Extraction:**
   - The robot uses *CNN-based feature extraction* to identify the kitchen counter, the ingredients, and the tools.
   - It also applies *pose and keypoint detection* to track precise hand movements, such as how the person exactly grips the whisk or the angle of the hand when pouring.
2. **Domain Bridging:**
   - The robot applies *image translation* to make the human's hand look like the robot's gripper in the video.
   - It also uses *context translation* to adapt the skill learned in a brightly lit human kitchen to be successfully executed in the robot's dimly lit, specialized lab environment, effectively adapting the action plan to the novel context.

**Summary & Limitations**

Image and context translation provide complementary approaches to bridging domain gaps between training data and real-world execution. Image translation focuses on visual alignment, making human demonstrations look like robot observations, while context translation captures the underlying task structure and relationships that enable generalization across varying environments and embodiments [33].

However, methods still lack at precise action alignment, translation artifacts, and managing mismatches between embodiment alignment [33].

### 6.2.2 Reinforcement Learning

Reinforcement Learning (RL) enables robots to acquire manipulation skills through trial-and-error interaction guided by reward signals. By continuously interacting with the environment, RL can optimize policies for complex and long-horizon tasks that are difficult to solve with purely supervised approaches. However, RL typically requires large amounts of interaction data and carefully designed reward functions, making real-world robotic training expensive and time-consuming [33]. While simulation environments can alleviate this issue, they often require substantial engineering effort and suffer from sim-to-real transfer challenges. Leveraging information extracted from large-scale video data can significantly improve the data efficiency, robustness, and generalization capabilities of RL systems [73].

In the context of learning from videos, RL approaches often combine exploratory learning with structured information extracted from human demonstrations [33]. Rather than forming strictly separated categories, the following approaches represent complementary strategies that can be integrated within broader RL frameworks.

- **Visual RL with Feature Extraction**

  Visual RL with feature extraction focuses on deriving meaningful visual representations from pretrained models on video data and using them to guide reinforcement learning. Video demonstrations are transformed into structured training signals, such as object trajectories, motion representations, latent embeddings, or dense reward functions, which help RL agents learn more efficiently from limited interaction data [104, 92, 164]. These approaches reduce the need for manually engineered rewards and improve policy learning by grounding exploration in semantically meaningful visual features.

- **Structured and Hierarchical RL**

  Structured and hierarchical RL approaches decompose complex tasks into reusable skills, subtasks, or high-level goals. Instead of learning a single monolithic policy, the agent learns hierarchical representations that separate high-level planning from low-level control. Video data can be used to learn task embeddings, latent skills, or subgoal representations that enable multi-task learning, long-horizon reasoning, and improved generalization across manipulation tasks [142, 74, 21].

  In particular, hierarchical and multi-task RL frameworks enable the reuse of learned skills and improve generalization across diverse environments.

### 6.2.3 Imitation Learning

Imitation Learning is one of the promising directions for learning robot policies from videos. It enables robots to acquire skills by directly mimicking actions observed in human or robot demonstrations, without the need for explicit reward functions or extensive trial-and-error interaction. Imitation learning policies can be more sample-efficient and often have a better generalization than pure Reinforcement Learning policies, as they leverage the rich information contained in demonstrations to guide policy learning [33].

Imitation Learning is divided into the subcategories Bahavior Cloning and Learning from Videos / Observation, see Section 3.2.2.

Learning from Videos / Observation approaches can be formulated as an inverse Reinforcement Learning problem, where the goal is to infer the underlying reward function that explains the observed behavior in the video. Once this cost function is recovered, a policy can be optimised to minimise the predicted cost, thereby reproducing the expert's behaviour [33, 127]. Generative Adversarial Imitation from Observation (GAIfO) [127] employs a generative adversarial setup where a discriminator is trained to differentiate between state transitions coming from expert demonstrations and those generated by the robot's current policy. Simultaneously, the robot policy learns to produce transitions that the discriminator cannot distinguish from expert data, thereby matching the observed dynamics of the expert. This is achieved without any explicit action labels and offers a more scalable alternative to traditional IRL [33].

In addition to these main categories, several complementary approaches have been developed to further enhance the capabilities of imitation learning from videos [33].

**Meta- & Few-Shot Imitation Learning**

Meta- and Few-Shot Imitation Learning approaches focuses on improving generalization capabilities by training on diverse tasks so that robots can quickly adapt to novel tasks from only a handful of demonstrations [33, 36, 114].

`Few-shot Learning` refers to a broad approach in which models can make accurate predictions using only a small number of examples. It is a subcategory of n-shot learning, and includes One-shot learning (Learning from a single example) and Zero-shot learning (no examples needed). In practice, few-shot learning often leverages techniques such as meta-learning, transfer learning, data augmentation, or data generation methods, like (e.g. using generative AI). Combinations of these approaches are also common [39, 114].

`Meta-learning`, also known as "learning to learn", describes the development of models that can achieve strong performance from a small amount of data or with very few training steps, thereby adapting efficiently to new tasks. More formally, Meta-Learning is a approach that aims to improve the learning ability of models by enabling them to learn from prior experiences and transfer that knowledge to new tasks. The main goal is to adapt the learning ability from humans. These algorithms are typically trained on a wide variety of different tasks, that the model can learn how to quickly adapt to new tasks [33, 39, 114].

**Methods**

Meta-learning approaches like Model-Agnostic Meta-Learning (MAML) have been applied to imitation learning, where policies are pre-trained on diverse tasks so that a single new demonstration suffices for adaptation [153].

**Cross-Domain & Human-to-Robot Transfer**

Cross-Domain and Human-to-Robot Transfer methods aim to achieve robust transfer from human video demonstrations to robot execution, addressing the challenges of embodiment differences and domain shift between human and robot data [33]. These approaches typically learn domain-invariant representations, intermediate action representations, or apply domain adaptation techniques to bridge the gap between human demonstrations and robot execution.

**Methods.**

Multi-step task learning from human videos [41] localises human-demonstrated actions within supplemental videos and combines behavioural cloning (BC), inverse RL (IRL), and RL for accelerated policy refinement.

Watch, Try, Learn (WTL) [160] combines visual meta-learning with binary feedback (success/failure). A robot first watches a human demonstration, then attempts to perform the task. After each attempt, it receives only a simple success or failure signal (not a detailed reward). Using meta-learning, the robot rapidly updates its policy across a small number of such trials to refine its behavior.

Visual Entity Graphs (VEGs) [111] models spatial and temporal relations between objects and actions using graph-based representations. By explicitly modelling how entities interact over time, the graph provides a structured, domain-invariant representation. This enables a robot to learn from a single demonstration: the graph abstracts away low-level visual differences between human and robot embodiments, allowing direct transfer of action-relevant relationships to the robot's own environment.

### 6.2.4 Hybrid Approaches

Hybrid approaches combine reinforcement learning (RL), imitation learning (IL), and complementary techniques to overcome the limitations of relying on a single learning paradigm. The central motivation is to integrate the sample efficiency and structured guidance of imitation learning with the exploration and optimization strengths of reinforcement learning [33] [98]. By leveraging multiple training strategies simultaneously, hybrid methods aim to improve data efficiency, robustness, and generalization in complex robotic manipulation tasks.

A common strategy is to use Imitation Learning for policy initialization, followed by RL-based fine-tuning to improve long-horizon performance and adapt policies through interaction with the environment [98, 88]. Other approaches incorporate *Residual RL*, where reinforcement learning refines or corrects imperfect demonstration trajectories, enabling adaptation to different robot embodiments or environmental conditions [38].

Hybrid methods are also widely used for *domain adaptation*. By exploiting semantic priors, representation learning [88, 98], or adversarial objectives [38], these approaches enable robots to transfer skills across different domains, viewpoints, and embodiments, even when demonstrations are unaligned or originate from third-person videos [116, 98, 90, 3]. Adversail Learning = More recent work further *combines RL and IL with control-theoretic methods* such as Model Predictive Control (MPC), improving trajectory optimization, safety, and real-world deployment robustness [44]. MPC is an advanced control strategy that uses a Dynamics Model $p(o_{t+1} \mid o_{t-k:t}, a_t)$, to predict future behavior [102].

**Discover the usage of Video Data for RL Knowledge Modalities**

Reinforcement Learning (RL) can benefit from knowledge extracted from large-scale video data in multiple ways. In the following, we analyze how information obtained from videos can be incorporated into different RL Knowledge Modalities in order to improve learning efficiency and overall performance:

- Policy
- Dynamics Model
- Reward Function
- Value Function

**Policies**

Representations can be pretrained from video data and used downstream whilst training a policy on robot data. Jointly train on video, robot, image and text data to predict actions to predict robot actions. A policy trained to output an action representation which conditions an action-decoding policy trained on robot data. specific instantiation: hierarchical setup where the action representation is a video and the decoder is (often) an inverse dynamics model (IDM)

The core idea is to learn useful representations, actions, or abstractions from video data $D_{video}$ to improve robot policies $\pi(a_t \mid o_t)$.

**a) Representation Transfer**

Large-scale video data can be leveraged to pretrain visual models that learn semantically meaningful representations of environments, objects, and task dynamics. Instead of learning directly from limited robot interaction data, the agent first acquires general visual priors from diverse video datasets. These pretrained representations can subsequently be transferred to downstream policy learning, where they improve both data efficiency and generalization capabilities.

In this setting, Video Foundation Models (see Section 5.1), are typically trained using self-supervised or unsupervised objectives on large collections of videos. The learned visual encoder captures high-level features such as object permanence, motion patterns, spatial relationships, and temporal consistency. During downstream reinforcement

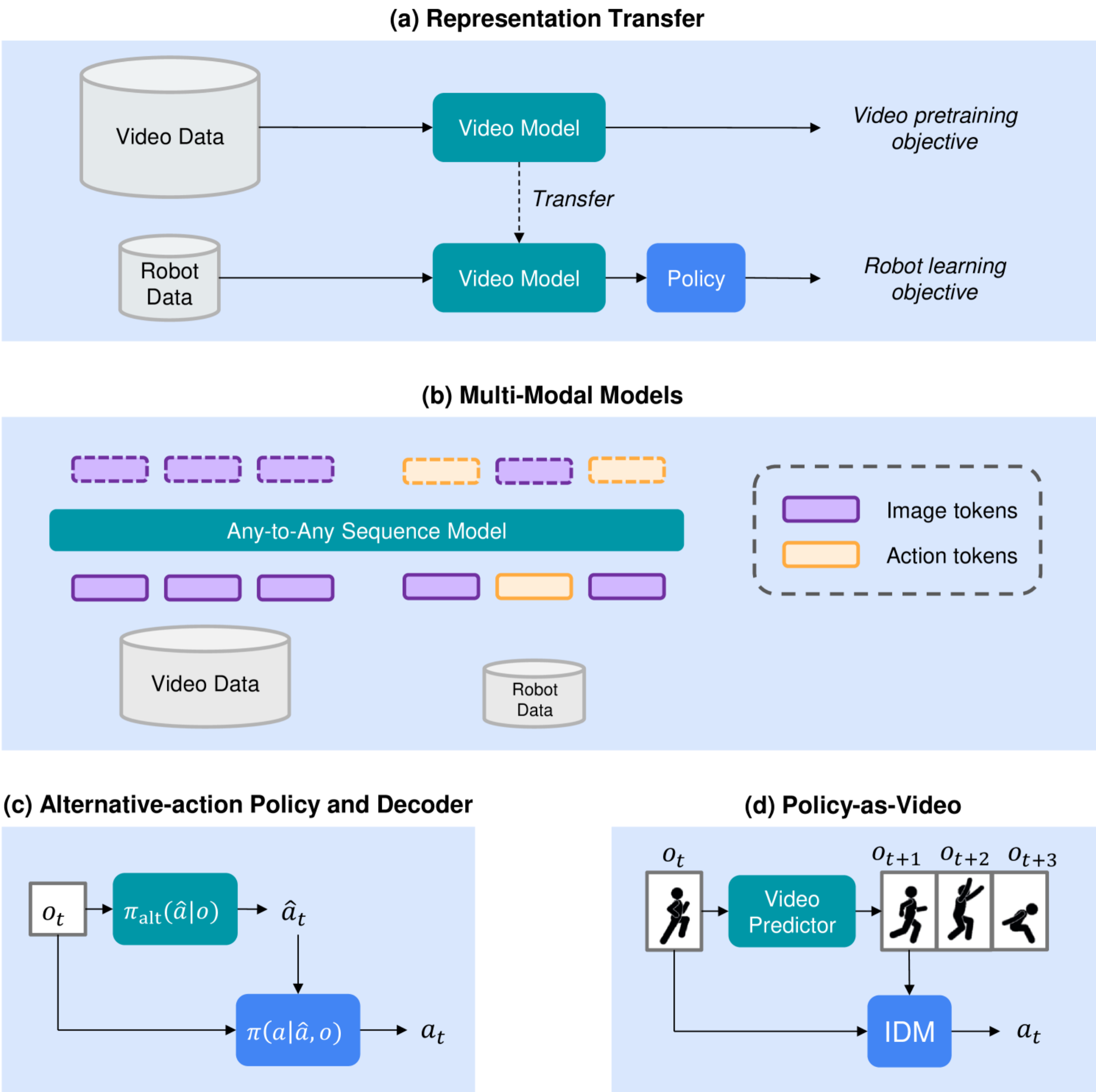



**Figure 6.2:** Learning Policies from Video [73].

or imitation learning, the pretrained encoder is either frozen or fine-tuned while the policy is optimized on robot-specific data.

The main advantage of representation transfer is that the policy no longer needs to learn low-level visual features from scratch. Instead, it can focus on task-specific decision making, substantially reducing the amount of robot interaction required for effective learning. Furthermore, pretrained representations often generalize better across environments, viewpoints, and object variations, making policies more robust in real-world settings.

Several large-scale studies have demonstrated the effectiveness of transferring representations learned from video data to robotic policy learning [**120**, **246**, **328**, **41**, **108**, **174**].

**c) Alternative action Policy $\pi_{alt}(\hat{a}_t|o_t)$ and Decoder $\pi(a_t|\hat{a}_t, o_t)$**

Alternative action Policy $\pi_{alt}(\hat{a}_t|o_t)$:

An Alternative-action policy $\pi_{alt}(\hat{a}_t|o_t, c_t)$ learns to predict an Alternative Action Representation $\hat{a}_t \in \hat{A}$ based on observation $o$ and some other conditional information $c$, rather than directly outputting low-level robot actions. For Further Information about Alternative Action Representations see Section 5.2.

Alternative-action policies are typically trained on large-scale internet video data to gather diverse behavioral patterns and general knowledge about the world and physical interactions.

A common training procedure assumes that the video data contains approximately expert behaviour. The videos are first labelled with alternative-action representations, after which the alternative-action policy $\pi_{alt}(\hat{a}_t \mid o_t)$ is trained using supervised behaviour cloning objectives [105, 19, 133, 30, 13, 138, 11]. The learned policy can then either be used directly for representation transfer or to condition a downstream action decoder $\pi(a_t \mid \hat{a}_t, o_t)$ trained on robot interaction data.

When the behaviour contained in the video data is noisy or suboptimal, additional conditioning signals can improve learning. In particular, language conditioning [30, 138] and goal conditioning [133] help constrain the learned behaviour towards task-relevant objectives. Offline reinforcement learning may provide another mechanism for leveraging imperfect demonstrations, although this direction remains relatively underexplored in the current literature.

Several recent approaches instantiate alternative-action policies using pretrained vision-language or large language models on internet-scale data. For example, Yang et al. [148], Du et al. [30], and Mu et al. [78] finetuned VLMs or LLMs to produce language-based action representations. In contrast, Black et al. [13] train an image-editing diffusion model that transforms the current observation into a predicted subgoal image, which serves as the alternative-action representation.

Alternative Action Decoder $\pi(a_t|\hat{a}_t, o_t)$:

A separate Action Decoder $\pi(a_t|\hat{a}_t, o_t)$ can subsequently be trained on robot data to map these Alternative Action Representations $\hat{a}_t \in \hat{A}$ to real robot actions commands $a_t$. This approach to define an Alternative Action Policy and Decoder effectively decouples the learning of high-level action representations from low-level control, allowing the policy to leverage rich information from video data while still grounding its outputs in robot-compatible actions.

There exist two approaches to obtain the decoder: `Learning the decoder.` In this setting, the robot dataset $D_{robot}$ is augmented with alternative action representations, resulting in tuples of the form $(o_t, \hat{a}_t, a_t, o_{t+1})$. The decoder $\pi(a_t \mid \hat{a}_t, o_t)$ can then be trained using supervised learning objectives to predict low-level robot actions conditioned on both the current observation and the alternative action representation.

Other Approaches to learn a decoder include:

- Instead of keeping the alternative-action policy fixed, some approaches jointly optimize the compositional policy
$$\pi(a_t \mid \pi_{alt}(\hat{a}_t \mid o_t), o_t)$$
using online reinforcement learning, enabling end-to-end training between high-level planning and low-level control [105].
- [**jain2024vid2robot**] learns a direct mapping from human demonstration videos to robot actions from paired human-video and robot-trajectory data in an end-to-end setting.

`Manually defining the decoder.` Instead of learning the decoder from data, the decoder can be manually crafted using explicit engineered geometric or kinematic mappings. Manually defined decoders reduce the need for paired robot supervision and can incorporate strong inductive biases about the robot embodiment.

Examples of manually defined decoders include:

- Inferring low-level robot actions from optical flow patterns in the video, which capture the direction and magnitude of motion in the scene.
- Mapping robot actions to visual arrows, which projecting future robot pose as on-screen hints and subsequently decoding them back into executable control commands.
- Retargeting human hand poses or body motions to equivalent robot joint commands.

Example Workflow with $\hat{A}$ = Policy-as-video

In this workflow, generated videos serve directly as the policy output. The procedure consists of two steps:

1. A **language-conditioned video predictor** $p(\tau \mid o_{t-k:t}, c)$, generates a future video sequence that imagines successful task completion given the current observation and a language instruction. $\tau$ represents a generated video trajectory and $c$ some conditional information, like language commands or other context.

2. An **action-decoding inverse-dynamics model (IDM)** $p^{-1}(a_t \mid o_t, o_{t+1})$ is used to infer low-level robot actions $a_t$ from these generated video frames. The IDM is trained on the robot dataset $D_{\text{robot}}$.

Potential improvements of this compositional approach include:

- Compositional alignment between video, language, and the IDM actions within a multi-modal approach, where each modality informs the others.
- Improving the action decoding model by incorporating goal-conditioned behavior cloning.
- Add additional conditioning modalities of Alternative Action Representations $\hat{A}$ to the IDM to improve the predicted low-level robot actions.

**Summary & Limitations**

Overall, hybrid approaches provide a flexible framework that combines the strengths of multiple learning paradigms. This synergy enables robots to learn robust skills from limited demonstrations, adapt policies online, and generalize more effectively across tasks and environments.

**Dynamics Models**

Dynamic models $p(o_{t+1}|o_t, a_t)$ aim to determine how the environment evolves over time by predicting future observations based on the current state and the next action to be taken. To learn dynamics models from video data, the core idea is to leverage the rich temporal and physical information contained in videos to train models that can capture the underlying dynamics of the environment.

One common approach is to first pretrain a Video Prediction Model $p(o_{t+1} \mid o_t)$ on large-scale video datasets $D_{video}$ in order to learn general temporal dynamics and physical interactions of the environment. The pretrained model is subsequently adapted into a robot dynamics model $p(o_{t+1} \mid o_t, a_t)$ using robot interaction data $D_{robot}$, incorporating the influence of robot actions on future environment states. This transfer enables the model to leverage broad visual world knowledge learned from internet videos while grounding the predictions in robot-specific control dynamics.

Another approach is the usage of Alternative Action Representations by training an alternative-action video predictor $p_{alt}(o_{t+1} \mid o_t, \hat{a}_t)$, where $\hat{a}_t$ denotes high-level actions or latent behavioural representations extracted from video data. The learned alternative-action representations can subsequently condition a downstream action decoder trained on robot data. Conditioning on an alternative action space $\hat{A}$ provides greater control over generated trajectories and often enables mappings between abstract video-derived actions and executable robot actions.

A further direction jointly pretrains dynamics models on both video and robot datasets. Recent approaches such as UniSim extend this paradigm by conditioning dynamics models on temporally extended actions across multiple modalities, including robot motor commands, language instructions, and actions extracted from camera motions and or other video-derived action representations.

LfV video predictors have been conditioned on:

- language-as-actions [306, 68]
- single-step latent actions [221, 40]
- goal images [67]
- future grasp-location and post-grasp waypoint affordance information [180]
- motion information [319, 279]

306 condition on several different action-spaces, including robot actions, language actions, and camera motions

**Adaptation of video predictors to robot dynamics models**

Adaptation of video predictors to robot dynamics models often involves a fine-tuning step using robot interaction data $D_{robot}$. One challange is that naive finetuning on robot data result in the erasure of knowledge acquired during large-scale video pretraining. To mitigate this issue, [232] preserve pretrained knowledge by stacking an action-conditioned prediction model on top of the pretrained video predictor instead of directly modifying the entire model. **180** first applies video-based conditioning mechanisms during pre-training and subsequently incorporates robot-specific action conditioning in the adaptation phase. **305** demonstrate that the score function of a large pretrained video diffusion model can guide the generation process of a smaller task-specific video model, enabling efficient specialization while retaining knowledge from large-scale pretraining. Adaptation can furthermore involve learning mappings from alternative action representations $\hat{A}$ to executable robot actions [**221**].

The resulting dynamics models can then be used for

- **Generate synthetic training data** The model can generate synthetic training data by simulating how the environment evolves in response to different actions. This allows for data augmentation and policy training without requiring real-world robot interaction.
- **Differentiable Simulator:** By enabling backpropagation through model-generated rollouts, the dynamics model allows policies to be trained end-to-end using gradient-based optimization. This is especially useful for learning complex behaviors efficiently.
- **Planning and Control:** The dynamics model can be used for model-based control and trajectory planning, allowing the robot to anticipate the consequences of its actions and make informed decisions. Examples are here Tree search and Model Predictive Control (MPC). MPC optimizes a trajectory over a short horizon and replans frequently. Systematically simulates and evaluates many possible action

sequences, often over a longer or variable horizon, by branching at each decision point.

- as part of a larger policy learning framework, where the dynamics model provides a predictive component that informs policy updates, value estimation, or hierarchical decision making.

**Evaluation of Model-Generated Trajectories.** To assess the quality of simulated trajectories, it is necessary to evaluate them using a value or reward function. The reward function can be learned from reward-labeled robot data or even from video data, enabling the model to assign meaningful scores to predicted outcomes and guide decision-making.

### Discussion

Large pretrained video foundation models demonstrate strong generalization capabilities and enable dynamics models to learn useful world knowledge even from non-expert internet video data. In particular, diffusion-based video models have shown promising performance for long-horizon prediction and planning tasks.

Future research directions include scalable Alternative Action $\hat{A}$ conditioning methods, hierarchical world models that combine video-level abstractions with robot-level control, and integrating learned video predictors with classical physics simulators. Another promising avenue is the generation of synthetic robot training data and trajectory rollouts using large pretrained video generation models. An underexplored direction is to combine standard analytic or physical simulation [267, 175] with video prediction simulation systems [306] to combine the physical accuracy and controllability of analytic simulators with the strong visual realism and planning capabilities of video predictors. Inspired by synthetic data practices in other domains [287, 150], future research could leverage foundational video predictors to create synthetic video rollouts [306, 40] or to expand and enhance existing dataset diversity through video editing techniques [316].

Overall, video-based dynamics models provide a scalable framework for bridging visual world understanding and robot control through large-scale video pretraining, action conditioning, and model-based planning.

### Reward Functions

Reward functions are a central component of reinforcement learning, as they define the objective that guides policy optimization. However, manually designing reward functions for open, complex and unstructured robotic environments is difficult, often task-specific and hard to scale. To address this limitation, Learning from Video (LfV) approaches aim to automatically extract visual reward functions directly from video data. The core idea is to transform robot transition tuples $(o_t, a_t, o_{t+1})$ into reward-annotated tuples $(o_t, a_t, r_t, o_{t+1})$, enabling their use in both online and offline reinforcement learning settings.

Video-Language Model as Reward Functions

One common approach is to use Video-Language Models (VidLMs) to derive reward signals from language descriptions and video observations. [73] identify two categories of methods:

- **Video-text similarity:** Dual-encoder architectures embed videos and natural language instructions into a shared representation space. The reward is defined as the similarity between the robot observation and the language-conditioned task representation in the embedding space.
- **Visual Question Answering (VQA):** Video-to-text models answer questions about the current state of the task, such as whether the task has been completed or estimating the current task progress. The answers are then used to provide dense reward signals that guide policy learning. [] used VQA with images as input to provide rewards.

Video Prediction Models as Reward Functions

Video Prediction Models $p(o_{t+1}|o_t)$ learn to predict future observations based on the current visual state of the environment. Such models can be converted into reward functions by measuring how likely the next performed robot action, as observed by it's camera, is under the pretrained video predictor. In this setting, the robot receives a high reward when the actual next frame from the robot's camera closely matches the predicted future observations of the model. This encourages the robot to produce behaviors whose visual dynamics resemble those observed in the video data used during pretraining. To improve scalability across diverse and non-expert internet videos, recent approaches additionally condition the video predictor on language instructions, enabling task-specific reward generation for a broad range of behaviors.

Visual Representational Similarity to a Reference as Reward Functions

This approaches define the reward as the similarity between the current robot observation and a reference observation, such as goal images or demonstration videos. In this setting, deep visual representations are used to compare the semantic similarity between observations rather than relying on raw pixel comparisons.

Several methods employ time-contrastive learning objectives, which provide implicit estimates of the temporal distance between two observations and therefore encode task progress information about the current robot observation and the visual reference. Meanful visual representations are needed to address the distribution shift between human demonstrations and robot observations, like

- viewpoint-invariant representations,
- embodiment-invariant representations, and
- object-centric representations.

Other methods apply pose retargeting techniques to map human motions to robot-compatible embodiments before computing the reward signal.

Potential-based Shaping with Value Functions

Dense reward signals can be obtained from pretrained value functions $V(o_t, c)$ or $Q(o_t, a_t, c)$, where $c$ represents the goal condition of the task. The value function estimates how much progress has been made toward achieving the goal represented by $c$ from the current observation $o_t$. Such value functions can be learned from video demonstrations using either Temporal-Difference (TD) learning or Time-Contrastive learning, see Section 6.2.4.

Once a value function $V(o_t, c)$ is learned, it can be used to generate a dense reward signal via potential-based reward shaping:

$$r_t = V(o_{t+1}, c) - V(o_t, c)$$

This reward is positive when the robot moves toward states with higher predicted success (i.e., makes progress toward the goal) and negative when it regresses. This formulation provides smooth and dense optimization signals, which can significantly improve the stability and efficiency of downstream reinforcement learning. The principle can be applied to action-value functions $Q(o_t, a_t, c)$ too.

Generative Adversarial Imitation Learning (GAIL)

GAIL trains a policy (the robot's behavior) and a discriminator simultaneously in an adversarial game.

- The **discriminator** is trained to distinguish between state-action pairs coming from the expert demonstrations (e.g., videos) and those generated by the current robot policy.
- The **robot policy** is trained to produce state-action pairs that fool the discriminator into thinking they came from the expert.

In this setup, the discriminator's output serves as a **learned reward signal**: when the robot's behavior looks similar to the expert, the discriminator is uncertain and gives a high score; when the behavior looks different, the discriminator confidently classifies it as fake and gives a low score. The result is that the robot learns to maximize this score and match the distribution of behaviors observed in the video demonstrations, without ever receiving an explicit reward function.

Since human demonstrations and robot observations often differ significantly in viewpoint, embodiment, and visual appearance, these approaches typically require invariant feature representations to reduce the impact of distribution shift between video and robot domains.

Other Reward Extraction Methods

Additional methods train a task classifier on task-labelled video datasets to provide downstream reward signals for reinforcement learning. Other approaches estimate the number of remaining steps-to-completion in a video and use this quantity as a proxy signal for task progress.

Further work encourages similarity to video-derived behavioral priors, such as pretrained video policies or human affordance distributions, allowing the robot to align its behavior with patterns extracted from large-scale video demonstrations.

**Downstream Usage of LfV Reward Functions**

Video-pretrained reward functions can either be transferred zero-shot to downstream robotic tasks or further finetuned on robot data $D_{robot}$. The extracted rewards may serve as the sole optimization objective for reinforcement learning, or act as auxiliary shaping and exploration rewards in combination with sparse rewards. In some approaches, the reward functions are additionally used to pretrain policies before subsequent finetuning the reward model on manually defined task rewards.

**Summary**

Compared to directly learning policies or dynamics models, reward functions are often easier to extract from large-scale video data, since they only require estimating task progress or behavioral similarity rather than generating complete action sequences. Recent advances in pretrained foundation models, particularly video-language models and video prediction architectures, have significantly improved the quality and scalability of visual reward extraction.

Future research directions include combining LfV reward functions with other video-based knowledge modalities, such as finetuning policies through online Reinforcement Learning using the learned reward functions, as well as using LfV rewards to augment reward annotations in datasets. Another promising avenue is the use of video-derived rewards for safety monitoring and constraint estimation. Furthermore, extracting dense reward signals from video-to-text models through Reinforcement Learning from AI Feedback (RL-AI-F) represents an emerging direction.

Overall, video-based reward functions provide a scalable mechanism for translating visual demonstrations into optimization signals for reinforcement learning, reducing the need for manually engineered reward design in robotic systems.

**Value Functions**

Value functions play a central role in reinforcement learning, as they estimate the expected return or task progress from a given state or state-action pair. In the context of Learning from Video (LfV), recent approaches investigate how value-function-like models can be pretrained on large-scale video data, with the goal of capturing task-relevant progress signals that can be transferred to downstream robotic control. State-action-value functions $Q(o_t, a_t)$ are preferable to state-value functions $V(o_t)$, because they offer more informative guidance for action selection and policy improvement.

**Pretraining: Temporal-Difference Learning**

One common approach is to pretrain value functions using the Temporal-Difference (TD) Learning. TD learning requires often access to actions, rewards, and goal annotations, which are typically not available in raw video data.

To handle missing action labels, value functions are often conditioned on Alternative Action Representations $\hat{A}$, resulting in models of the form $V(o_t, \hat{a}_t)$.

To adress the lack of reward and goal annotations, LfV reward methods described in the Section 6.2.4 can be used. Some approaches leverage object labels together with off-the-shelf object detectors to infer task progress and goal states from observations. In manipulation tasks, object detectors can identify relevant objects, locations or the robot gripper in each frame. By tracking these objects over time, the system can estimate task progress and automatically derive reward signals. For example, decreasing distance between the gripper and an object may indicate approach behavior, while object movement together with gripper interaction can signal successful grasping or task completion. This enables the generation of goal and reward annotations without requiring manually labeled demonstrations. Furthermore, video-to-text models can provide semantic goal descriptions in natural language, enabling value functions to be conditioned on textual task objectives.

**Pretraining: Time-Contrastive Learning**

Value functions can be learned through Time-Contrastive Learning. Time-contrastive Learning approaches learn task progress directly from the temporal structure of videos. The model is trained such that observations temporally closer to the goal condition $c$ obtain more similar representations, while temporally distant observations are separated in the embedding space. For example, a task can be specified via a set of goal images. The model learns that observations close in time to the goal should have high value, while observations far from the goal (in time) should have low value. As a result, the learned representation implicitly captures progression toward task completion and can be used to derive dense reward signals from raw video demonstrations.

Other Approaches discover the use of:

- *Quasimetric functions*, to learn an asymmetric distance measure that better captures task progress. For example, going from "object on table" to "object in box" is meaningful progress, while the reverse transition should not necessarily have the same distance.
- *Temporal cycle-consistency objectives* enforce that temporal correspondences between frames remain consistent across videos. If frame A in one video corresponds to frame B in another demonstration, the mapping should also hold when traced back, producing temporally aligned representations across demonstrations.

These approaches do not require action labels, making them particularly suitable for large-scale video pretraining. However, they typically require explicit goal images or reference observations at training and inference time.

#### Pretraining: Heuristic and Weak Supervision

Another line of work constructs value functions using heuristic supervision signals extracted from video data. A common strategy is to regress the number of remaining timesteps until task completion, effectively learning a time-to-go metric under the assumption of expert behavior in the provided task demonstration. Other approaches finetune vision-language models to predict such heuristic progress signals.

Other methods additionally identify key states in demonstration videos and use them to improve value estimation, although these approaches often still rely on some form of reward or success annotation in the training data.

#### Downstream Usage

Video-pretrained value functions can be used in several ways in downstream robotic learning. If the value function is explicitly action-conditioned $V(o_t, a_t)$ or alternative-action conditioned $V(o_t, \hat{a}_t)$ with a mapping to robot actions, it can be directly integrated into standard value-based reinforcement learning or planning frameworks. When combined with a learned or given dynamics model $p(o_{t+1} \mid o_t, a_t)$, unconditioned value functions $V(o_t)$ can also be used for tree-search based planning.

In addition, pretrained value functions can serve as initialization for downstream policy or value networks, improving training efficiency in robot domains. They are also commonly used for potential-based reward shaping, where dense rewards are defined as $r_t = V(o_{t+1}) - V(o_t)$ to guide exploration in sparse-reward settings. Finally, value functions can accelerate TD learning in downstream tasks by serving as bootstrap targets in Bellman updates.

#### Summary

Research on learning value functions directly from video remains relatively scarce, mainly due to missing action, reward, and goal labels, as well as the multi-modal nature of human behavior in video data. TD learning from video shares similar challenges with offline RL, but existing solutions from that literature (e.g., conservative regularization or latent action inference) offer potential fixes. Despite these difficulties, TD.learning of value functions from large-scale human video is increasingly seen as a promising direction for real-world robotics, enabling reward-free progress estimation and goal-conditioned policy learning.

### 6.2.5 Multi-Modal Approaches & VLA Models

Multi-modal robotic learning aims to integrate several modalities, including visual observations, language instructions, robot states, tactile information, and robot actions, into a unified learning framework. The main objective is to improve task understanding, robustness, and generalization across diverse environments and manipulation tasks.

These approaches are considered an important step toward general-purpose robotic intelligence.

Multi-modal models that explicitly include robot actions are often called Vision-Language-Action (VLA) models, which jointly model visual perception, language understanding, and robotic control. VLA models extend Vision-Language Models by incorporating robot actions as an additional modality, enabling end-to-end policy learning from observations and natural language instructions. Their goal is to learn general-purpose robot policies capable of executing complex tasks across different environments.

Another objective of VLA models is to bridge the semantic gap between high-level multimodal inputs and low-level robotic control. To achieve this, they learn shared representations across modalities such as Language, Vision (images, videos), Robot States and Actions and sometimes other modalities, like proprioception or depth. Through large-scale training on multimodal datasets, VLA models align semantic understanding with motor behavior, allowing robots to generalize across diverse tasks and environments.

### General VLA Architectural Overview

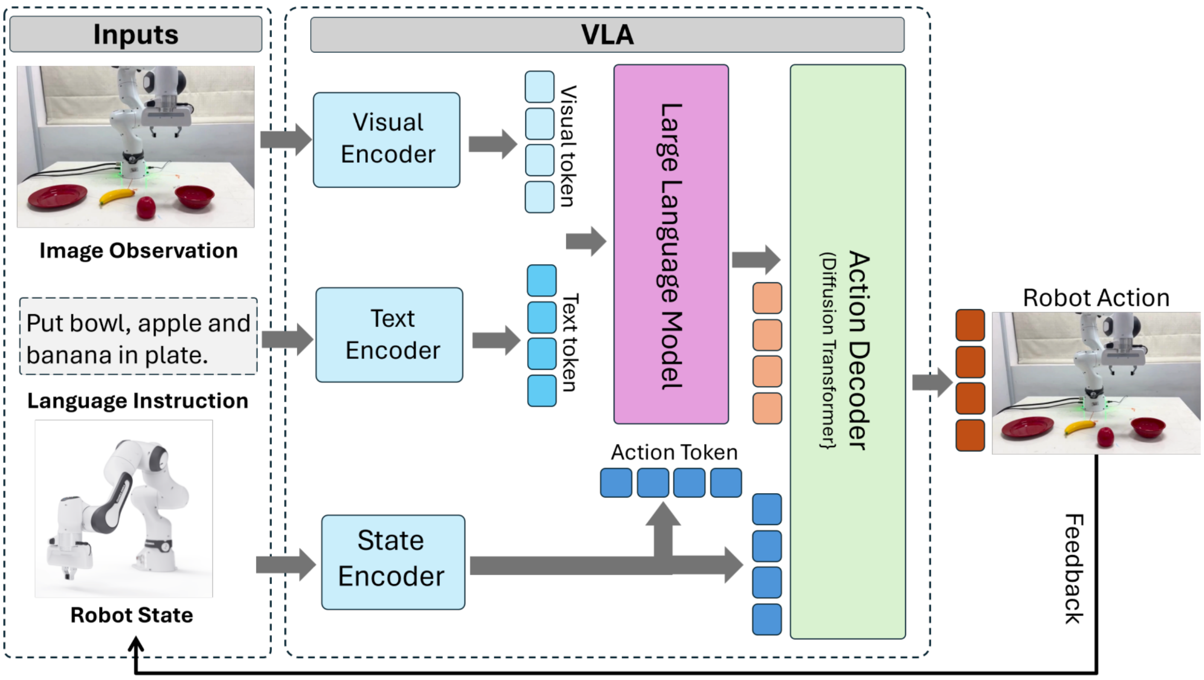



**Figure 6.3:** General Vision-Language-Action Architecture [?].

Figure 6.3 illustrates the general architecture of a Vision-Language-Action (VLA) model for robotic manipulation. Such systems typically process three main inputs: a visual observation of the scene, a natural language instruction, and the robot's internal state information. These inputs are encoded separately using Visual, Language, and State Encoders to obtain modality-specific feature representations.

The resulting embeddings are then fused within a large language model (LLM) or multimodal transformer, which performs high-level semantic reasoning and generates a semantic representation of the task. Finally, this representation is combined with robot state features and passed to an Action Decoder, which predicts the sequence of robot actions or trajectories required to execute the commanded task. The Action Decoder can take various forms and is here illustrated as Diffusion Policy, which directly generates actions based on the fused representation.

Technically, these systems combine advances in deep representation learning, multimodal alignment, and decision making, often leveraging transformer-based architectures trained with large-scale supervised or autoregressive objectives.

The mind map in Figure 6.4 shows the main categories of Vision Encoders, Language Encoders, and Action Decoders used in modern Vision-Language-Action (VLA) models. The listed architectures represent common design choices for multimodal perception, semantic reasoning, and robotic action generation in current large-scale robotic systems.

Some models appear under multiple encoder categories because they employ hybrid architectures that combine several pretrained backbones within a unified framework, to improve visual representation quality, grounding capabilities, and downstream manipulation performance.

**Main VLA Architectural Paradigms**

Vision-Language-Action (VLA) models encompass a broad range of architectural designs that differ in how perception, language understanding, and robotic control are integrated. Learning paradigms also vary and include supervised learning, self-supervised learning, and reinforcement learning, which can be combined in various ways. [**?**] categorizes VLA architectures into three main classes: sensorimotor models, world models, and affordance-based models. Each class represents a different strategy for grounding multimodal inputs into robotic actions and has its own architectures.

- **Sensorimotor Models.** Sensorimotor Models jointly learn representations of visual observations, language instructions, and robot actions within a unified policy model. These architectures directly map images and textual commands to robot actions and may employ either flat end-to-end policies or hierarchical structures that separate high-level reasoning from low-level control.
- **World Models.** Beyond direct policy learning, several alternative VLA paradigms have emerged. World-model-based approaches learn the temporal dynamics of the environment by predicting future visual or latent states conditioned on actions and language inputs. These predicted future trajectories can then guide planning and action generation.
- **Affordance-Based Models.** Another direction are Affordance-Based Models, which predict task-relevant affordances or interaction regions from visual and language inputs before generating corresponding robot actions.

Together, these architectural strategies illustrate the ongoing shift from task-specific

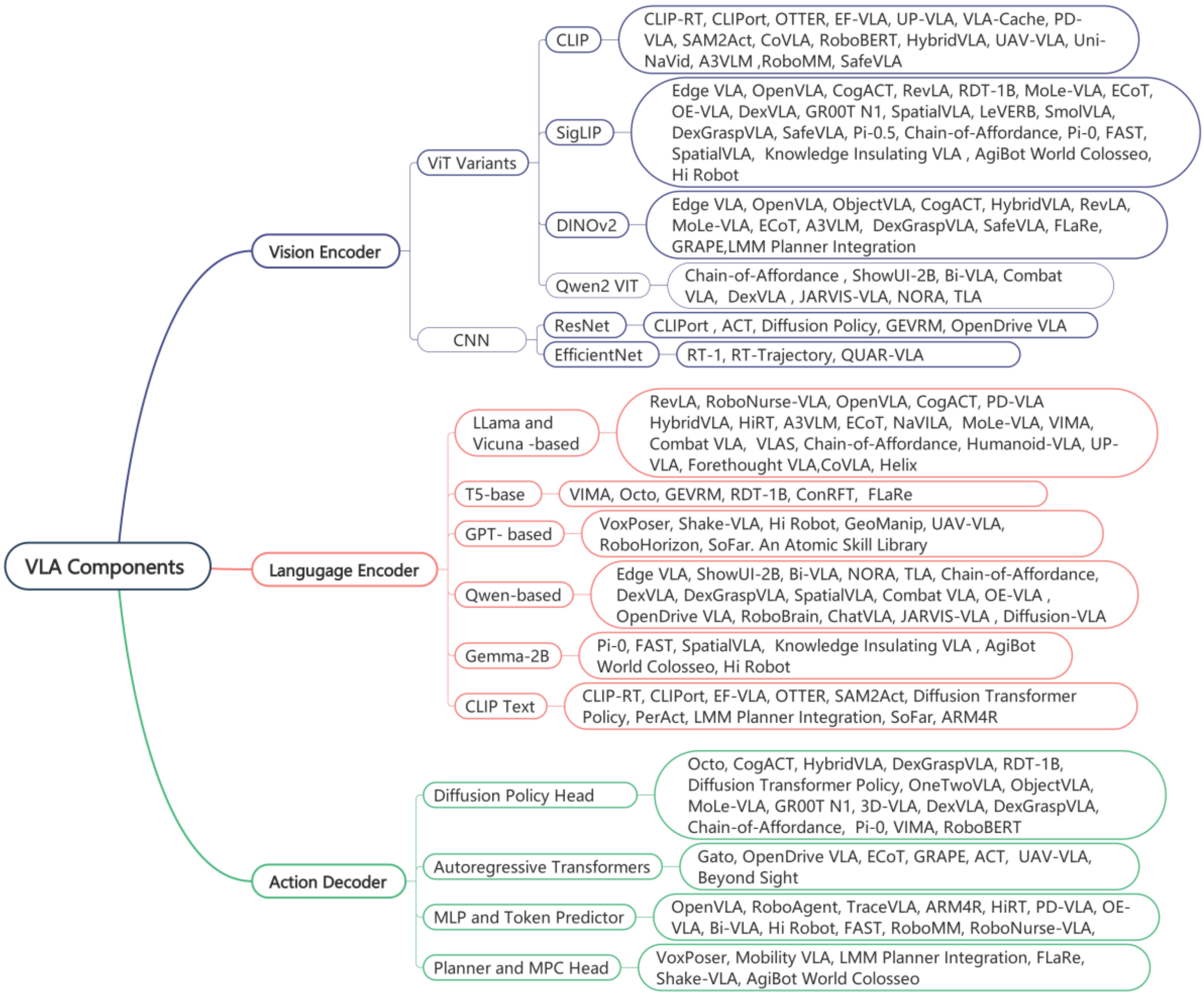



**Figure 6.4:** Vision-Language-Action Components Overview [?].

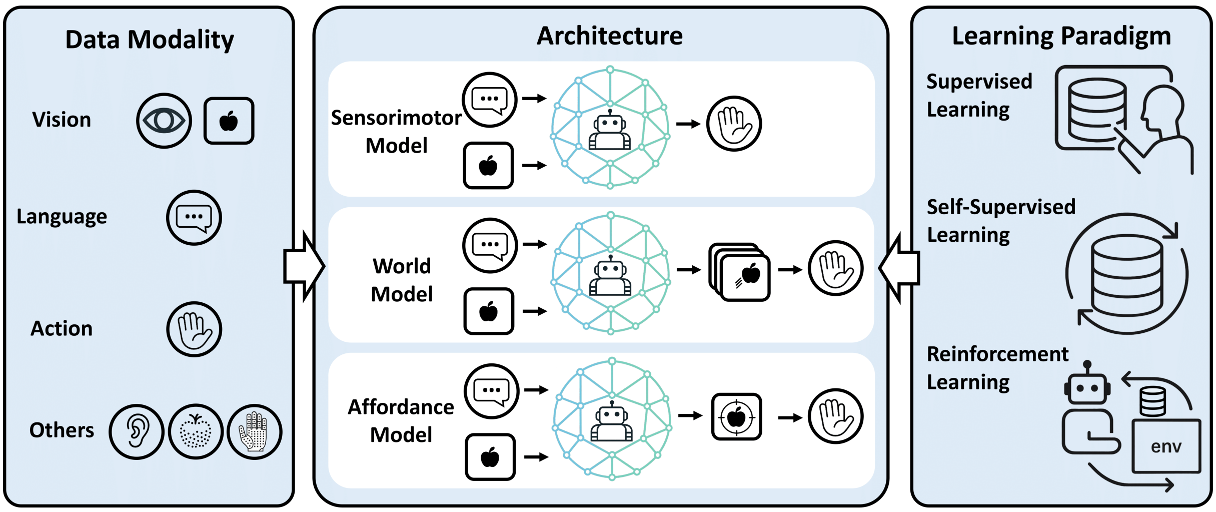



**Figure 6.5:** Vision-Language-Action Modality Architecture and Learning Paradigm [?].

robotic policies toward more general-purpose, multimodal robotic intelligence systems. VLAs represent the recent frontier in robotic intelligence by jointly integrating visual perception, natural language understanding, and robot action generation into a unified framework.

### 6.2.6 Comparison of approaches

# Chapter 7

# Challenges and Future Directions

Despite significant progress in imitation learning and video-based robotic learning, several fundamental challenges still limit the deployment of robust and generalizable robotic systems. Current methods often struggle with data availability, computational scalability, domain transfer, and efficient policy learning. At the same time, emerging architectures such as diffusion models, world models, and causally-aware systems provide promising directions for future research.

## 7.1 Data

The performance of modern Robotic models strongly depends on the availability, diversity, and quality of demonstration data. However, collecting large-scale robotic datasets remains expensive and time-consuming, particularly when expert annotations or robot demonstrations are required. Existing datasets are often domain-specific, weakly balanced, or limited to narrow task distributions, reducing the ability of models to generalize to unseen environments, objects, or robot embodiments.

One promising approach is the use of internet videos, which provide a vast amount of diverse visual demonstrations at a significantly lower collection cost than robot-specific datasets. However, such data often lacks precise action annotations and task labels.

To address this limitation, weakly supervised learning methods utilize coarse or incomplete labels, such as video captions, task descriptions, or automatically generated annotations, instead of relying on expensive frame-by-frame expert labeling. Similarly, self-supervised learning objectives enable models to learn useful visual and temporal representations directly from raw video data. Common objectives include predicting future video frames, reconstructing masked observations, or learning representations that remain consistent across temporally related frames. These approaches allow robots to exploit large amounts of unlabeled data while reducing the dependence on manual annotation.

Another challenge is the efficient acquisition of high-quality labels. Active learning addresses this issue by allowing a model to strategically select the most informative or uncertain samples for annotation rather than labeling entire datasets uniformly. For example, a robot may request human annotations only for demonstrations in which its predictions are uncertain, thereby reducing annotation costs while maximizing the benefit of labeled data.

Beyond passive observation, future imitation learning systems may also integrate active physical trials, where robots interact directly with their environment to refine policies learned from demonstrations. Such interaction-based learning combines the strengths of imitation learning and reinforcement learning, enabling robots to adapt to environmental variations, recover from failures, and acquire knowledge that is difficult to infer from observation alone.

Another important challenge is sample efficiency. Many high-capacity imitation learning and RL methods still require large amounts of demonstrations or interaction data. Current approaches therefore investigate meta-learning, contrastive representation learning, and one-shot or few-shot imitation learning to reduce data requirements. Goal-conditioned RL and Representation Learning have also shown improved data efficiency, although robust learning from weakly-labeled or noisy demonstrations remains an open problem.

Future research must focus not only on improving generalization, but also on enabling rapid adaptation to new tasks, robots, and environments. Achieving this level of adaptability is considered a key requirement for scalable and general-purpose robotic intelligence.

## 7.2 Computional Costs

The increasing adoption of large-scale transformer architectures and multimodal foundation models has significantly improved robotic perception and policy learning. However, these advances come at the cost of high computational and memory requirements.

State-of-the-art Vision-Language-Action models often require substantial GPU resources during both training and inference. This creates major deployment challenges for real-world robotic systems, especially on embedded or resource-constrained platforms where real-time execution is required.

Future work must therefore address efficient scaling strategies for both data and model architectures. Potential directions include for example lightweight transformer designs, modular architectures and model compression techniques.

## 7.3 Domain Shift

A persistent challenge in imitation learning is the domain shift between human demonstrations and robot execution. In particular, transferring skills from humans to robots is difficult because of differences in appearance, motion dynamics of task execution, embodiment, environment and action spaces.

Current approaches attempt to learn domain-invariant or domain-adaptive representations using hybrid and multi-modal architectures, image translation techniques, or context-aware feature learning. Methods such as adversarial learning, keypoint-based transfer, and CycleGAN-based translation can partially reduce visual discrepancies between domains. However, translation artifacts, inaccurate pose estimation, and misaligned action representations still limit robust skill transfer.

Future research will likely focus on more robust sim-to-real transfer methods, continual learning strategies during inference, and advanced domain adaptation techniques. Multi-modal learning approaches that integrate additional sensory modalities, such as force or tactile feedback, may further reduce the dependence on purely visual transfer. In addition, causal reasoning and few-shot adaptation methods could improve the transfer of manipulation skills across different robots and environments. Furthermore, researchers should investigate in attention mechanisms, unsupervised techniques, and transformer-based architectures.

## 7.4 Emerging techniques and architectures

Many traditional imitation learning approaches are designed for single tasks and often rely on a single input modality, such as visual observations or expert demonstrations. Such methods often struggle to generalize across diverse tasks, environments and robot embodiments, limiting their applicability in real-world robotic systems. Furthermore, learning complex manipulation behaviors from demonstrations requires models capable of producing high-dimensional outputs, such as complete action trajectories or future visual observations, rather than single action predictions.

To address these limitations, recent research increasingly focuses on multi-task and multi-modal learning architectures that can leverage information from diverse data sources and transfer knowledge across tasks. In particular, diffusion models and world models have emerged as two promising architectural paradigms.

### 7.4.1 Diffusion models

A fundamental challenge in imitation learning is the inherently multi-modal nature of many robotic tasks. For a given observation, there is often no single optimal action. For example, an object can be grasped from multiple valid directions, and a drawer can be opened using different trajectories. Traditional approaches typically learn a deterministic mapping from observations to actions and are commonly optimized using regression objectives such as mean squared error. As a result, they tend to predict an average of multiple valid behaviors, which may lead to suboptimal or even infeasible actions.

Diffusion models offer a promising solution to this problem by learning the full distribution of successful behaviors rather than a single deterministic output. Originally developed as generative models for image synthesis, they are particularly well suited to robotics due to their ability to represent complex, high-dimensional, and multi-modal distributions. Instead of producing a single averaged action, diffusion models can generate diverse and physically plausible action sequences that reflect the multiple valid solutions inherent in many manipulation tasks. This is especially important in real-world environments, where uncertainty, noise, and variability make multiple strategies equally valid.

In robotics, this idea has been adopted in diffusion policies, where robot actions are generated directly from observations—typically visual inputs—through an iterative denoising process. Rather than predicting a single action at once, the policy is trained to denoise an action trajectory conditioned on the current state of the environment. During inference, this trajectory is progressively refined over multiple denoising steps, resulting in temporally coherent and precise behavior. By capturing the full distribution of expert demonstrations, diffusion policies enable more robust and flexible action generation compared to deterministic regression-based approaches.

Diffusion-based policies have also emerged as a powerful framework for continuous robot action generation. Unlike earlier approaches based on discrete action prediction, diffusion models can generate smooth and temporally consistent trajectories while capturing multimodal action distributions. This enables more expressive and stable robot behavior, particularly for dexterous manipulation and long-horizon tasks.

Building on these advantages, diffusion-based policies have emerged as a powerful framework for imitation learning and represent a promising direction for future research.

### 7.4.2 World models

While diffusion policies focus on directly generating actions, an alternative approach is to learn a predictive model of the environment itself. A key limitation of many current imitation learning systems is that they react to observations without explicitly reasoning about the future consequences of their actions. This often leads to inefficient learning and limited long-term planning capabilities.

World models address this challenge by learning the transition dynamics of the environment. Formally, a world model approximates the probability distribution

$$p(s_{t+1} \mid s_t, a_t)$$

which describes how the environment evolves from the current state $s_t$ to the next state $s_{t+1}$ after executing an action $a_t$. By repeatedly applying this learned transition model, a robot can predict future states and forecast complete trajectories before interacting with the real world.

This capability enables a form of internal planning often referred to as *imagination*. Instead of relying solely on trial-and-error interactions, the robot can simulate the consequences of different action sequences within its learned world model and select actions that maximize task success. Consequently, world models provide the foundation for model-based planning and can be significantly more sample-efficient than model-free approaches, which typically require large amounts of real-world interaction data.

Recent advances increasingly combine world models with generative architectures. In particular, video prediction models can be interpreted as world models that learn to generate future visual observations conditioned on actions.

For example, modern systems employ diffusion transformers to generate high-resolution, long-horizon videos of robotic manipulation tasks that act as learned world models and interactive simulators. This enables policy evaluation, model-based planning, and skill learning to be performed entirely within the generative model, thereby mitigating the cost, safety risks, and labor associated with extensive real-world robot rollouts [202].

As generative modeling techniques continue to advance, world models are expected to become a central component of future robotic systems by combining perception, prediction, planning, and control within a unified framework.

### 7.4.3 Integration of causal reasoning

A current limitation of imitation learning systems is their weak ability to perform causal reasoning and high-level policy abstraction from video data. Most existing approaches focus on pattern recognition, goal inference, or directly imitating demonstrations. However, they do not explicitly learn the causal structure behind actions and outcomes, meaning they do not understand why a certain action leads to a result.

The main issue is that neural networks typically learn statistical correlations from data, but correlation does not necessarily imply causation. This leads to models that may rely on irrelevant features in the environment. For example, a robot might incorrectly associate the color of an object with the force needed to lift it, instead of learning that mass is the true determining factor. The goal of causal reasoning is therefore to capture the underlying cause-and-effect structure of the environment.

Such causal models are considered a foundation for better generalization. They enable robots to act more robustly in changing environments, adapt to novel situations, and perform counterfactual reasoning, meaning they can reason about what would have happened if a different action had been taken (e.g., pushing instead of grasping an object). Overall, causality supports more efficient, robust, and intelligent decision-making and improves long-term action planning.

To achieve this, recent approaches explore causal inference and model-based reasoning, as well as neuro-symbolic methods. Neuro-symbolic systems combine the pattern recognition ability of neural networks with the logical reasoning of symbolic AI, allowing for more structured and explainable decision-making.

#### Methods for Causal Discovery

A key research goal is causal discovery, which aims to infer the causal structure of the environment, typically represented as a Directed Acyclic Graph (DAG), which is a directed graph that does not contain any cycles. Causal discovery methods can be broadly categorized into observational and interventional approaches.

### Observational Approaches

Observational approaches learn causal graphs from passively collected video data. It is divided into two main categories:

- **Constraint-based methods** prune graph edges from a fully connected graph based on conditional independence tests. They identify causal relationships by determining which variables are conditionally independent given others.
- **Score-based methods** define a scoring function that evaluates how well a graph structure fits the data and search for the best graph according to this score.

**Example: Visual Causal Discovery Network (V-CDN).** The V-CDN architecture comprises three main components. First, a perception module processes raw images to produce an unsupervised, temporally stable keypoint representation of the observed objects. These keypoints act as the nodes of a causal graph. Second, an inference module analyzes the movement of these keypoints over a short video sequence to deduce a latent causal structure, identifying which keypoints influence one another. Third, a dynamics module—implemented as a graph neural network—takes the inferred causal graph as input and learns to forecast the system's future state.

### Interventional Approaches

Interventional Approaches are the gold standard. They involve actively manipulating variables to observe effects, providing unambiguous evidence.

**Example: SCALE (Skills from CAusal LEarning).** SCALE focuses on learning manipulation skills that transfer across different environments. Instead of relying on fixed context variables, it uses a simulator as a causal reasoning engine to perform targeted interventions. The method identifies the smallest possible set of context variables (e.g., object mass, position, size) that truly matter the causal structure by checking whether changing each variable alters the task outcome. The learned skill is then conditioned exclusively on this compact set of causal variables, leading to better generalization.

**Example: Causal Robot Discovery (CRD) framework.** Building on SCALE, CRD involves building an initial causal model from passive observation. Conditional independence tests are used to analyze the model and identify the most uncertain or unreliable links and is an online learning process. Based on this uncertainty, the robot plans and executes its next set of interventions, creating an efficient, self-improving feedback loop that actively seeks the most informative data. For example, if the model is uncertain about whether an object's mass affects the success of a lifting task, the robot can perform interventions by changing the object's mass and observing the outcome.

**Causal Representation Learning**

A key bottleneck for causal discovery methods is that they operate on predefined Features, yet real-world robotic systems receive only raw sensory data such as images, which are translated into a high-dimensional learned feature space. Causal Representation Learning addresses this gap by learning to map high-dimensional observations onto a low-dimensional latent space whose dimensions correspond to the independent causal mechanisms of the environment. An ideal Causal Representation Feature Space, factors such as object identity, pose, lighting, and background are disentangled into separate, independently controllable latent variables or dimensions. Such a representation is inherently more compositional and generalizable: a robot could reason about and intervene on each factor without affecting the others.

Although still an emerging field, CRL holds promise for enabling more sample-efficient and robust imitation learning, particularly when transferring policies from simulation to the real world.

## 7.5 Evaluation Metrics and Benchmarks

Another major challenge is the lack of standardized evaluation metrics and benchmarking protocols for imitation learning and video-based robot learning. Current evaluations are often task-specific and rely on custom experimental setups, making fair comparisons between methods difficult.

Existing benchmarks primarily focus on task success rates, while aspects such as robustness, generalization, sim-to-real transfer, and adaptability are often insufficiently evaluated. Moreover, the design of a benchmark strongly influences research directions by implicitly prioritizing certain capabilities over others. Table 7.1 summarizes a three prominent benchmarks and their principal focus.

| **Benchmark** | **Focus** | **# of Tasks** | **Key metrics** |
|---|---|---|---|
| CALVIN | Long-horizon, language-conditioned, compositional tasks | 34 | Long-horizon multi-task language-conditioned control success |
| RLBench | Few-shot, meta- and multi-task learning; broad skill acquisition | 100 | Few-shot task success |
| RoboTube | Learning from human demonstrations; human-to-robot transfer | 50 | Policy success rate in RT-sim; sim-to-real transfer success |

**Table 7.1:** Prominent benchmarks and their focus

**The causal gap.** An important limitation of current benchmarks is that they primarily reward correlational, task-level success. They do not explicitly test whether an agent has learned causal relationships that govern object interactions and task dynamics. As a result, an agent may reach high success rates by exploiting statistical regularities in the training data rather than by acquiring a robust causal model. For example, a policy might learn that pushing a particular red button is followed by a light turning on, without understanding the causal mechanism. If the button's function were altered, an object's physical properties were changed, or an unobserved confounding factor were introduced, a correlational policy will often fail.

**Towards causal evaluation protocols.** To close the causal gap [**?**] propose evaluation protocols inspired by the Minimal Video Pairs (MVP) idea [207], adapted to robotic control. MVP forces models to distinguish minimally different scenarios by changing a single causal detail. A potential protocol would proceed in two stages:

1. Train agents using the standard benchmark dataset and metrics (as today).
2. Evaluate robustness and causal understanding on a suite of interventional test tasks where underlying causal factors are systematically perturbed. Examples include changing object mass or friction, rewiring the causal relation between a switch and an effect, or introducing previously absent confounders (e.g., making a drawer stick).

Under this protocol a purely correlational policy that overfits the training statistics is expected to fail on the perturbed tasks, while a model that internalizes causal relationships should either adapt its policy to the new dynamics or detect the distributional shift and trigger safer, more informed failure recovery strategies. Quantitatively, such a benchmark would report both standard task success and robustness metrics, such as performance drop under interventions, adaptation speed, and diagnostic measures of causal inference.

Developing and releasing such interventional benchmarks will provide a concrete, quantitative way to measure the benefits of causal reasoning in robotic manipulation, and will help steer the field toward methods that generalize in meaningful, physically grounded ways.

# Chapter 8

# Discover Selected Approaches

In this chapter, we will explore some Imitation Learning approaches, that were applied to real robots to get a better practical understanding of the theoretical field.

## 8.1 Paper - Vid2Robot: End-to-end Video-conditioned Policy Learning with Cross-Attention Transformers

## 8.2 Paper - ORION: Vision-based Manipulation from Single Human Video with Open-World Object Graphs

## 8.3 Paper - Learning an Actionable Discrete Diffusion Policy via Large-Scale Actionless Video Pre-Training

## 8.4 Paper - Learning Video-Conditioned Policies for Unseen Manipulation Tasks

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